

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

PUBLICATION DATE **7.10.2020**

VERSION **v. 1.1**

RELATED SUPPORT

– TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

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Q – Summary of Local stakeholder consultation

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Q - Contact information of VPA Implementer (mandatory)

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KEY PROJECT INFORMATION

GS ID of Project	GS11343
Title of Project	GS10875 Namene Solar Lights Distribution Programme in Southern African Countries - Zimbabwe VPA 1 (GS 11343)
Time of First Submission Date	07/10/2021
Date of Design Certification	N/A
Version number of the VPA-DD	Version 1.4
Completion date of version	18/10/2022
Coordinating/managing entity	Namene Solar Lights Ltd.
VPA Implementer (s)	Namene Solar Lights Ltd.
Project Participants and any communities involved	Namene Solar Lights Ltd.
Host Country (ies)	Zimbabwe
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	
Methodology (ies) applied and version number	CDM small-scale methodology AMS-III AR version 7.0 - Substituting fossil fuel-based lighting with LED/CFL lighting systems
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A

Project Cycle:	☒ Regular ☐ Retroactive
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Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
1 No Poverty	Annual average expenditure savings on lighting	\$49 million (\$211 per participant household).	USD
3 Good Health and Well Being	Decreased indoor air pollution in households	80%	Households
4 Quality Education	Number of employees provided skills development training	27	Contracted persons
7 Affordable and Clean Energy	Households with access to clean and affordable solar lights	231,600	Households
8 Decent Work and Economic Growth	Number of persons hired under the project	27	Contracted persons
13 Climate Action	Average annual GHG Emission Reductions	42,605	tCO ₂ e

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

The Namene Solar Lights Distribution Programme Southern African Countries - Zimbabwe VPA 1 has been designed to provide clean sources of lighting to the population without access to electricity. The programme will do so by distributing ultra-affordable pico-solar lights to rural households that currently rely on the use of kerosene and other fuels for lighting in the provinces of Mashonaland East, Mashonaland West, and Masvingo.

Currently in Zimbabwe there are 8.5 million people without access to electricity, representing 59% of the country's population. This is exacerbated in rural households, where the percentage without electricity is 80%, meaning 1.2 million households are without energy access¹. While some progress has been achieved in recent years, the rate of electrification is unstable, unaided by an increase in population growth, particularly in rural areas.

This Voluntary Project Activity (VPA) is a private sector initiative led by Namene Solar Lights and being supported by private investment and underpinned by market-based environmental financing mechanisms that provide the speed and capacity needed to scale the programme in Zimbabwe. Therefore, this VPA is aimed to complement existing government-led programmes in the country for eradicating the use of fossil fuels for lighting and to increase the energy access.

A critical element required for mass market adoption and penetration of clean energy solutions at the Base of the Pyramid (BoP) is a breakthrough in the affordability of clean and safe pico-solar lights as a substitute for polluting, dangerous kerosene lamps and paraffin candles that only produce a low quality, dim light, or torches which require expensive batteries and result in waste that can pollute the environment with toxins during degradation.

¹ <https://data.worldbank.org/indicator/EG.ELC.ACCS.RU.ZS?locations=ZW>

This VPA is designed to achieve this rapid adoption and high penetration by providing affordable high-quality solar lights and using regional clusters, such as rural schools. The lights shall be distributed either via an internal team of the PD in the host country or through contracting local distribution partners (more details provided in section A.1.2). By tapping into the voluntary carbon markets to monetise the greenhouse gas (GHG) emission reductions achieved by the programme, the price of the solar lights will be subsidised so that every household can afford them. Over a short period of time, the use of solar lights will result in direct savings from expenditures on kerosene, candles, or batteries used for lighting. This in turn, will enable the households to take their first steps on the clean energy ladder.

The solar lights distributed as part of the VPA will be compliant with VeraSol² (formerly, the Lighting Global Pico PV Quality Standards), based initially on Namene's SM200 solar PV light³ that provides portability, reliability and high quality, demonstrated by extensive tests conducted and customers' feedback in other African countries.

This VPA is compliant with the host country's legal, environmental, ecological, and social regulations. The project activity (distribution of solar lights to households) does not require a license or permit in Zimbabwe. The method of distribution (such as through community centres, social events, and door to door) is legal and also culturally appropriate. The product (the Project Lamps) meets the internationally recognised VeraSol quality standard. The Project Lamps are in compliance with the country regulations and not listed in the Electricity (Inefficient Lighting Products Ban and Labelling) Regulations, 2017⁴. Since the project is not distributing or transmitting electricity, it is not subject to apply for a license to operate. The Appendix 1 presents the results from the safeguarding assessment and reference to the relevant laws and regulations (where applicable). Regarding waste management, to date Zimbabwe has no legislation or policy on electronic waste management. The Environmental Waste

² VeraSol is the evolution of Lighting Global Quality Assurance. For more information visit - <https://verasol.org/about-verasol>

³ For more information about the SM200 solar light and our company, visit www.namenesolar.com

⁴ Inefficient Lighting Products Ban and Labelling regulation, Zimbabwe 2017

Management Act (20:27)⁵ only prohibits the discharge of hazardous substances into the environment, but there is no specific legislation regulating batteries. The VPA-DD describes how the project complies with the local regulations (or lack of) and will address the electronic waste produce at the end of life of the Project Lamps in section B2 - Table 3, Paragraph 8d and Section D.1.

This VPA supports the implementation of solar energy technologies for off-grid applications, in particular solar lights. This VPA will implement the type 1: Solar Lamps under AMS-III.AR Version 7 "Substituting fossil fuel-based lighting with LED/CFL lighting systems".

A significant market penetration of pico-solar lights in the hands of families in Zimbabwe will reduce GHGs and climate forcing black carbon emissions generated from fossil fuel-based sources of lighting used by households in rural areas in Zimbabwe.

In addition, the project delivers significant social impacts and positive outcomes:

- Education – extra study hours for children that supports higher school grade attainment.
- Health – better air quality in homes, brighter light to avoid straining eyes and avoidance of a source of potentially life-threatening fires.
- Financial – saving money due to the avoidance of regular purchase of kerosene or paraffin candles for lighting.
- Road accidents – visibility of pedestrians or cyclists at dusk, dawn and night.
- Safety - Female safety at night.
- Energy Access - access to the first rung of the clean energy ladder.
- Employment - Opportunities for local employment to support deployment of 652,000 solar lights into rural communities. Extension of the working day due to accessing affordable bright light creates an opportunity to earn more income.

Consequently, the UN SDG impacts of this Project, spanning local rural community, provincial and national levels, are significant, broad and far reaching.

⁵ Environmental Management Act [Chapter 20:27]. | UNEP Law and Environment Assistance Platform

A.1.1. Eligibility of the project under approved PoA

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1	VPA location and project boundary	The geographical boundary within which the technologies are installed will be in Zimbabwe, within the geographical boundary outlined in Section A.4.4.	The location of the VPA is specified in section A.4.4, in which the CME states that the location is within Zimbabwe, as outlined in the POA-DD
2	Avoiding double counting of emission reductions	Each VPA will ensure double counting of emission reductions is avoided, through the unique identification of each unit distributed by the project with an identification number.	The identification system employed is outlined in the VPA DD in section B.2. Each unit distributed is marked permanently with a unique identification number (serial number and GS VPA ID or GS PoA ID) to avoid double counting. Personal information of the technology users (solar lights) is also gathered during distribution. These details are uploaded to the PoA database for record keeping, auditing and transparency.
3	Double counting of Programme of Activities	Each VPA will show that it is exclusive to the PoA and not registered as another project activity or VPA under another	This VPA is neither registered as a project activity with GS or any

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/ Supporting evidence for inclusion
		PoA. The VPA applies an identification system enabling unique identifying of each unit distributed or its users so that the double counting of emission reductions can be avoided.	other standard or as a VPA of another PoA. The appropriate registries (Gold Standard and CDM) can be accessed to demonstrate this. Each Project Lamp will be identified with a unique identification number to confirm they are part of this VPA or PoA.
4	Specifications of Technology/ measure	Each VPA will involve the distribution of solar lights to provide clean lighting services to the end users. The products distributed under each VPA shall be in compliance with the technical and certification requirements specified in Table 1 of the PoA and with the CDM small-scale methodology AMS-III.A.R. version 7.	This VPA will involve the distribution of solar lights to households where fossil fuel is commonly used for lighting and not connected to the electricity grid. The details of the technology are presented in section A.3 and the technical specifications and compliance with the PoA and with the methodology are presented in Section B.2. of this VPA-DD.
5	Start Date	Each VPA must not start prior to the commencement date of the PoA. The start date will be the date of which the first Project	The Start Date of the VPA is not prior to the commencement of the PoA i.e. 24/11/2020. The date of the first Project Lamp

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
		Lamps was distributed under the VPA.	distributed in the VPA is 01/03/2022.
6	Methodology	The PoA and each VPA will be in compliance with the CDM small-scale methodology AMS-III.A.R. version 7. The applicability criteria of the methodology is presented in section B.2. of the PoA-DD.	The applicability of the methodology is justified in Section B.2.
7	Additionality	The additionality will be determined at the PoA level. All VPAs are located in a Least Developed Country or landlocked developing countries and can thus be considered as deemed additional in line with the paragraph 4.1.9 of the Community Services Activity Requirements (Version 1.2). The Financial Additionality is not required to be proven at the time of Design Certification. Also, there is no need to demonstrate the ongoing financial need at Certification Renewal	This VPA is within Zimbabwe, an LLDC listed under section A.4.1.2 of the PoA-DD. Projects within these countries are deemed additional, as mentioned under section A.4.2.1 of the PoA-DD.
8	Non-diversion of ODA	In case that any of the VPA receives ODA, it is ensured that there is no diversion of ODA, i.e.	A signed declaration confirming that there is no diversion of ODA for each

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
		that no ODA is provided under the condition that all or part of the carbon credits have to be returned to the donor country/entity providing ODA.	VPA is attached with the PoA-DD and individual VPA-DDs.
9	Target group	Each VPA will involve the distribution of solar lights to households where fossil fuel is commonly used for lighting and are not connected to the electricity grid.	This VPA will involve the distribution of pico solar lights to households where fossil fuel is commonly used for lighting and not connected to the electricity grid. The monitoring database will confirm the distribution of Project Lamps to the target group
10	Sampling requirements	VPAs under the PoA will adhere to the sampling requirements in the CDM methodology AMS-III.A.R v7.	The VPA shall follow the monitoring plan described in the PoA-DD and as per the methodology.
11	Scale of the activity	The project activity will be a small-scale activity as per the CDM methodology requirements and in accordance with the GS4GG. Emission reductions achieved by each one of the activities considered under the PoA are limited to a maximum of 60,000	The total number of technologies to be installed by the VPA will be calculated to ensure the emission reductions remain within this limit. In case the VPA is found to exceed the applicable limit during any verification, the claimable emission reductions shall be capped

No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
		tonnes of CO ₂ e in any year of their crediting period.	at 60,000 tonnes CO ₂ per year.
12	Eligibility of Technologies	As per the CDM methodology AMS-III A.R. version 7, examples of technologies are LED lamps charged by: a) renewable energy systems, like solar PV b) standalone distributed generation system c) the grid; or d) a combination of the above methods.	The VPA will distribute Project Lamps that include LED lighting systems charged by renewable energy (solar PV).. Project Lamps will meet the methodology eligibility criteria.
13	Conditions to be met regarding SDG outcome assessment	SDG outcomes and the methods of monitoring these outcomes are defined in the PoA-DD in section A.4.	The VPA includes the SDG outcomes in the monitoring parameters and report on these in each monitoring report. This is described in detail in section B.6 of the VPA-DD.
14	Conditions to be met regarding safeguarding principles	The safeguarding principles are defined at VPA level as described in the PoA-DD Section E	The VPA includes the Safeguarding Principles in the monitoring parameters and report on these in each monitoring report. The safeguards assessment is done at the VPA level. This is described

No. Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
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in detail in section D.1 of the VPA-DD.

The VPA is eligible under the GS Community Services Activity Requirements version 1.2 as presented in the table below:

Table 2 – Eligibility of the VPA under the GS Community Services Requirements:

Eligibility criteria category	Eligibility Criterion – Required condition	Justification
1. Eligible project types	2.1.2) Projects shall lead to climate change mitigation and/or adaption by providing or improving access to services/resources at household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	By providing solar lights in off-grid rural communities, the project uses renewable energy (solar PV) to improve access to clean sources of lighting services at household and community level.
	2.1.3) In relation to the above all Projects shall therefore confirm to Gold Standard for the Global Goals Principles & Requirements (and associated documents)	The project conforms with GS4GG Principles and Requirements. The project is eligible under section 3.1.1 of the GS Community Services Activity Requirements as it follows an established Gold Standard approved methodology (AMS-III.A.R). Concerning point 3.1.1.5, the project does not support geoengineering or entail energy

		production from fossil fuels or nuclear. Rather it supports a switch away from polluting technologies to an emissions-free means of lighting
2. General eligibility criteria – Type of Project	(a) Renewable energy: Renewable energy types such as solar (photovoltaic and solar thermal electricity generation), tidal/wave, wind, hydropower, geothermal, waste to energy and renewable biomass that are connected to mini grid or off grid solutions for targeted users and/or applications.	The project activity involves 'off-grid renewable energy solutions for targeted users or applications' by providing solar lights to end users in communities with no access to the electricity grid.
2. General eligibility criteria - Project Area, Boundary and Scale	Project Area and Boundary shall be defined in line with the applicable Methodologies or Product Requirements	The project area and the geographical boundary of the VPA is the provinces of Mashonaland East, Mashonaland West, and Masvingo, in the Republic of Zimbabwe as outlined in Section A.4.4. below. The VPA is a small-scale project, issuing emission reductions to up to 60,000 tonnes of CO ₂ e per year, as per paragraph 10 of the small scale UNFCCC methodology AMS-III.A.R v.7.
2. General eligibility criteria - Legal ownership	Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with FPIC	Legal ownership of the Products generated under the GS certification this VPA is clearly defined and the rights to the carbon credits is transferred from the solar light end user to the CME. This process is described below in section A.3.

	requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided	
Community Services Activity Additional Eligibility Requirements		
Principle 3. Stakeholder Inclusivity	CSA projects shall have specific stakeholder consultation requirements for certain project types including, but not limited to, hydropower and projects using biomass resource as given in Annex A of this document).	The project activity involves 'off-grid renewable energy solutions for targeted users or applications' by providing solar lights to end users in communities with no access to the electricity grid. According to Annex A of the CSA, the additional eligibility criteria is not applicable to this project activity as it does not fall under any of 1-9 project types.
Principle 4 – Demonstration of Real Outcomes	4.1.5 New Projects may seek Certification and receive Issuance of Gold Standard Certified Impact Statements or Products for a maximum of two Design Certification Renewal Cycles i.e., a total of 15 years issuance. 4.1.6 Product Requirements with specific requirements for total issuance period shall supersede information in this document. 4.1.7 Design Certification Renewal in the case of CSA projects is mandatory every 5 years as per the Principles & Requirements. For the first renewal, CSA Projects are not required to reassess the Baseline Scenario.	The VPA may seek certification and receive GS issued products for up to a maximum of two renewal cycles (a total of 15 years). As per 4.1.7 Design Certification renewal for this CSA VPA will be conducted every 5 years as per the GS Principles and Requirements. For the first renewal, the VPA is not required to reassess the Baseline Scenario.
Principle 5 – Financial Additionality &	4.1.8 All projects seeking the issuance of Certified	The VPA is implemented in Zimbabwe, which is a Landlocked Developing Country (LLDC), as

Ongoing Financial Need.	Impact Statements and/or Products shall demonstrate Financial Additionality in accordance with the Principles & Requirements and the applicable Product requirements. 4.1.9 Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: (a) Positive list (Annex B of this document) (b) Projects located in LDC, SIDS, LLDC4 (c) Microscale projects 4.1.10 All CSA projects shall demonstrate Ongoing Financial Need as per the Principles & Requirements. 4.1.11 For stacked Gold Standard Certified Impacts the project developer shall demonstrate the impact of the additional revenue streams qualitatively (i.e. provide a definition of what further benefit the finance will bring).	per the UN official list. As per paragraph 4.1.9 of the CSA requirements, the VPA is deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification. The VPA is not planning to have any stacked GS Impact Statements.
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A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

Namene Solar Lights Ltd. (NSL) as the CME will be leading this VPA. NSL will be responsible for coordinated all aspects related to the Gold Standard Certification and is the legal owner of the products generated from the project and has the legal rights for the credits.

End users are informed of their carbon rights waiver when taking part in the Project during briefings prior to the subsidised sale of Project solar lights to the end user during the crediting periods. During distribution of Project Lamps a Project Instruction and Process Manual will be used by Local Distribution Partner(s) with explicit instructions and wording to be used during the briefings and the sale of the lights.

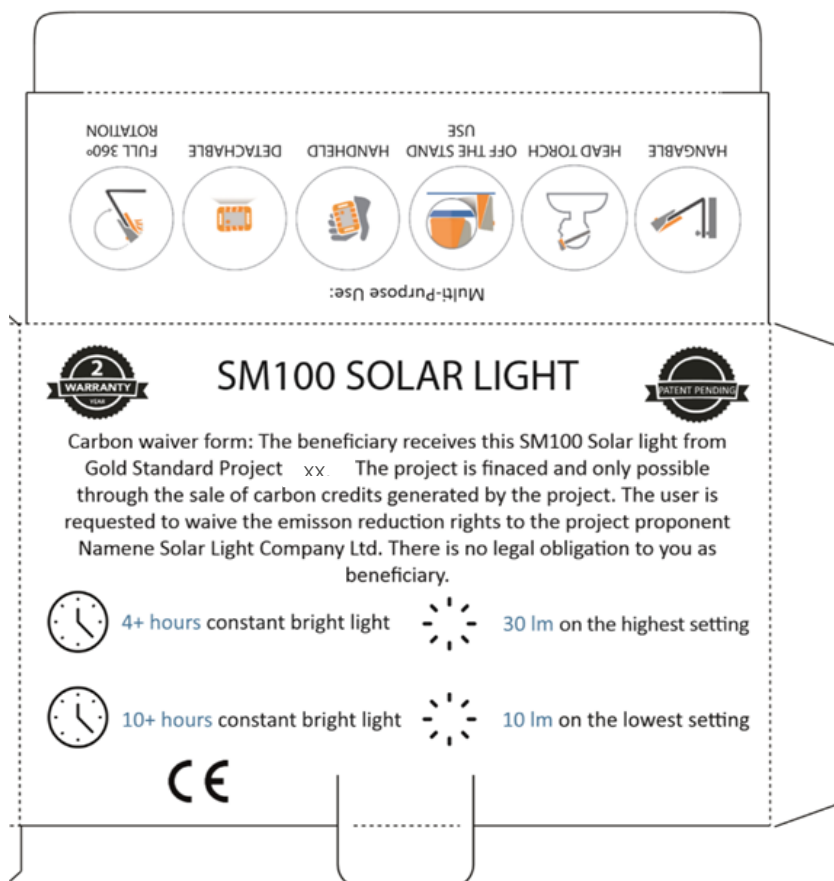
During the light sale process, end users will be taken by the Local Distribution Partner through the consent process regarding the carbon waiver before completing the sale.

In addition, the solar light product packaging also displays a printed carbon waiver statement as follows:

"Carbon waiver form: The beneficiary receives this solar light from Gold Standard Project VPA. The project is financed and only possible through the sale of carbon credits generated by the project. The user is requested to waive the emission reduction rights to Namene Solar Lights Ltd. There is no legal obligation to you as beneficiary."

The picture below shows a design of the light packaging for the project

Figure 1 – Example of the project Solar Light packaging showing carbon waiver form (Image rights – Namene Solar Lights Ltd.)



For Local Distribution Partner(s), a clause is included in a standardised Distribution Agreement to ensure full and uncontested legal ownership of carbon credits generated during the crediting periods.

A.2. Location of project

A.2.1 >> Host Country

The Republic of Zimbabwe

A.2.2>> Region/State/Province etc.

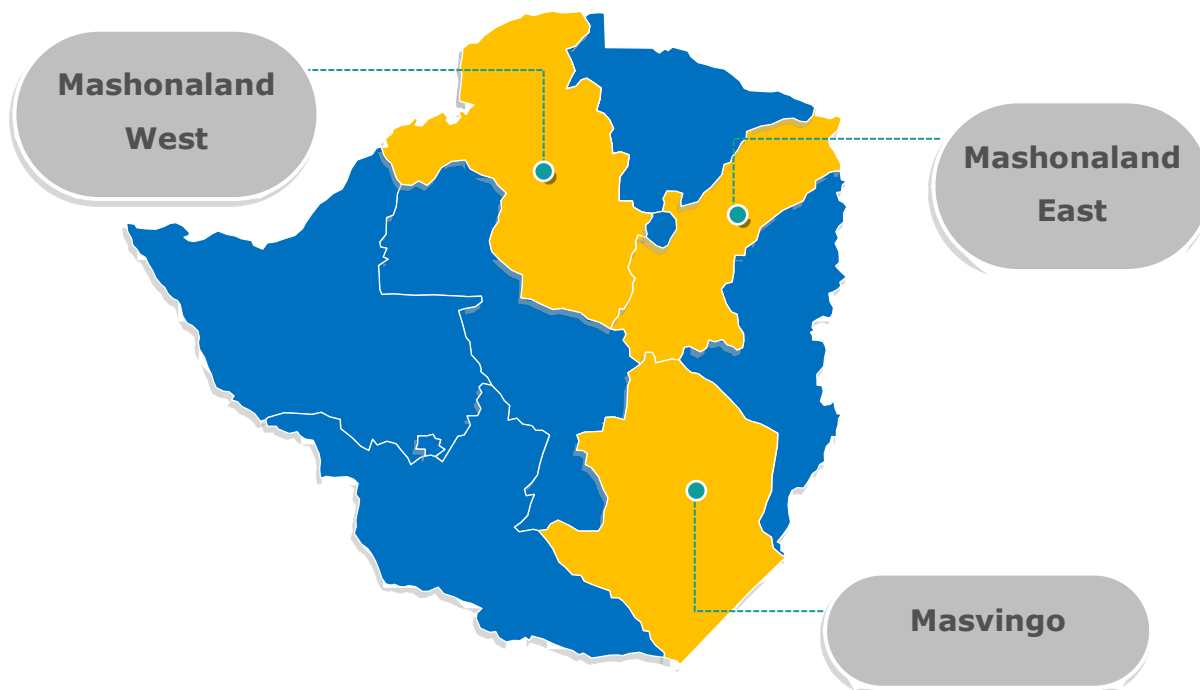
The Provinces of Mashonaland East, Mashonaland West, and Masvingo.

A.2.3 >>City/Town/Community etc

Rural areas and peri-urban areas with no access to grid electricity.

A.2.4 Physical/Geographical location

The geographical boundary of the VPA is the provinces of Mashonaland East, Mashonaland West, and Masvingo, in the Republic of Zimbabwe.



Due to the distributed nature of the technology deployed in this VPA (Solar lights), each Project Lamps will have a unique serial number and a VPA identifier or a PoA identifier, ensuring that the Project Lamps can be differentiated from other projects.

A.3. Technologies and/or measures

The measures implemented by this VPA will involve the distribution and use of solar lights, mainly in households that otherwise would commonly rely on fossil fuels or other polluting sources for lighting.

The activities implemented in this VPA are community service activities under the 'renewable energy' project type as per the Gold Standard approved "Community Services Activity Requirements" version 1.2. Specifically, the VPA involves 'off-grid renewable energy solutions for targeted users or applications'. The VPA will apply and meet the conditions defined in the small-scale CDM Methodologies AMS-III.A.R

version 7.0 and the Project Lamps will meet all the technical and methodological requirements, as described further in Section B.2, in Tables 3 and 4.

The VPA under this PoA will distribute solar lights in the provinces in Zimbabwe as defined in section A.4. There will be a maximum threshold of solar lights distributed in the VPA to meet the Gold Standard and CDM small scale methodology requirements. The Project lights will be distributed by the VPA implementer and/or distribution partners through awareness campaigns in rural communities, markets and local events, making the lights available to rural households that rely on fossil fuels for lighting. The Project Lights will be sold at a subsidised price (from the sale of carbon credits) to end users in order to allow all families to be able to afford them. The sale transaction including carbon waiver form and terms and conditions of the sale and warranty terms will be done by the sales representative of the VPA implementer and/or distribution partner and the records of the transaction kept in electronic or paper form.

As per section 4.1.3 of GS4GG Principles and Requirements (Version 1.2), the project types under this PoA are thus automatically eligible for Gold Standard Certification⁶. The detailed description how the VPAs meet the relevant eligibility criteria will be confirmed and described in detail in Section A.2.

Solar lights

This VPA will distribute pico-solar lights to end users (mainly households) that are not connected to the electricity grid in Zimbabwe, mainly in rural areas, where the access to electricity is more limited and the need to serve the Base of the Pyramid (BOP) is greater. The solar lights will reduce GHG emissions by displacing the use of kerosene, candles or other fossil fuels for lighting used in the absence of the project. The technologies employed in this VPA do not lead to GHG emissions.

⁶ As per the paragraph 4.1.3. of GS4GG Principles and Requirements (Version 1.2), a project type is automatically eligible for Gold Standard Certification if there are approved Gold Standard Activity Requirements and/or Gold Standard Impact Quantification Methodologies associated with it or as referenced in Gold Standard Product Requirements.

As an example, this VPA will start distribution with Namene Solar's SM200 multifunctional pico-solar light. As VPA may have progressive distribution schemes and new/other models may be introduced during the lifetime of the VPA.

Figure 1: Example of a solar light distributed by the VPA – SM200 Solar Light

SM200 Solar Light

Technical & functional data



All-in-one solar head torch, lamp and learning light



2 year warranty



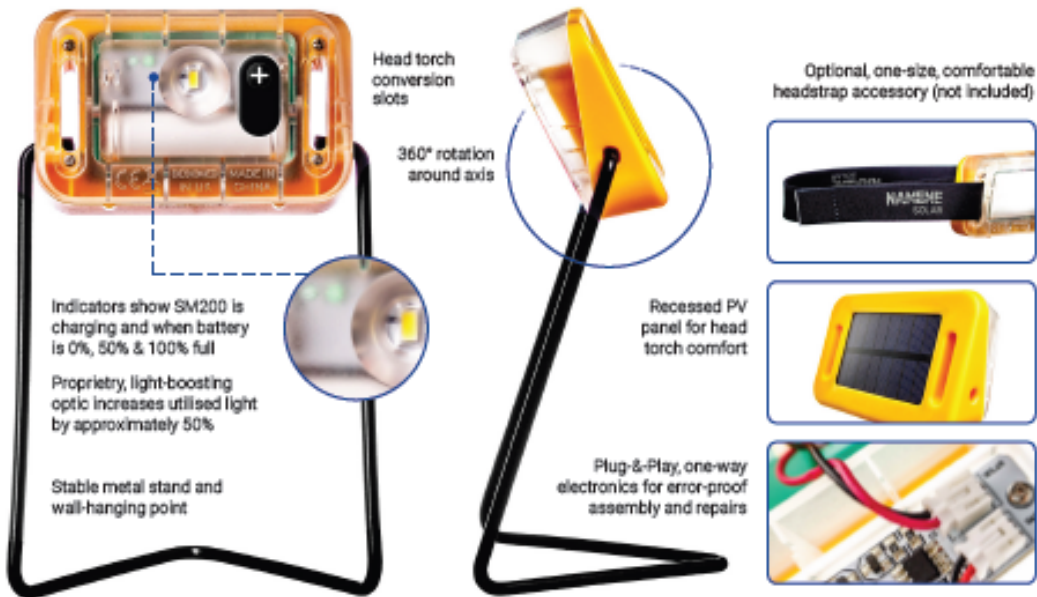
super bright mode (50 lm)



max. per daily charge (15 lm)



VeraSol certified



Main modes:

- 1. Bright
- 2. Brighter



Special modes:

- 1. Super Bright
- 2. Pulse

Mode	Brightness (lumens)	Solar run time (hours)	Full battery run time (hours)
Bright	15	12	21
Brighter	35	5	8.5
Super Bright	50	3	5.5



Solar PV panel:

4V, 75 mA, Polycrystalline, integrated mounting



Battery:

500 mAh, 3.2V, LiFePO₄



LED light source:

0.5 W cool-white LED



Ingress protection:

IP44 - resists solid objects over 1.0mm and water splashes from any direction



Tough build quality:

UV-resistant, impact-resistant & fire-retardant PC-ABS casing



Certifications:

Meets VeraSol IEC 62257-9-8 standard, CE, PVoC



Included as standard:

Positionable metal stand
Mounting slots for straps
Cardboard sleeve
Instructions & warranty info



Dimensions & Weight:

58 x 89 x 22 mm, 73 g (without stand),
165 x 105 x 74 mm, 125 g (with stand)



Logistics:

24 SM200 lights per box,
Box: 380 x 230 x 160 mm
1440 SM200 per master carton,
Pallet + master carton: 210 Kg,
1200 x 800 x 1180 mm

Namene Solar delivers solar power without limit. We want to bring everyone, everywhere access to clean, modern energy by connecting people to the sun to power their lives.

No solar light is too small to make a difference. Our SM200 follows our award-winning SM100, the popular head torch designed for off-grid. For more information, please visit:

www.namenesolar.com

Source and image rights: Namene Solar Light Ltd.

The distribution model for the solar lights will be conducted mainly through schools, and other similar aggregation centres to ensure an efficient distribution of the solar lights, as well as a robust process for raising awareness, providing support services and engagement with the communities benefited by the programme.

A.4. Scale of the project

The VPA is a small-scale activity as it will distribute solar lights that will not exceed the 60,000 tCO₂e annual emission reductions, as stated in the CDM small scale methodology AMS-III.AR. v7.0 and also in compliance with the small-scale definition by Gold Standard.

A.5. Funding sources of project

NSL provides the funding for the VPA and it is privately funded. There will be no diversion of ODA for this program or any of the proposed VPAs. A declaration of non-use of ODA has been completed and submitted.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

This VPA applies the CDM small-scale methodology AMS-III A.R. version 7.0 - Substituting fossil fuel-based lighting with LED/CFL lighting systems.

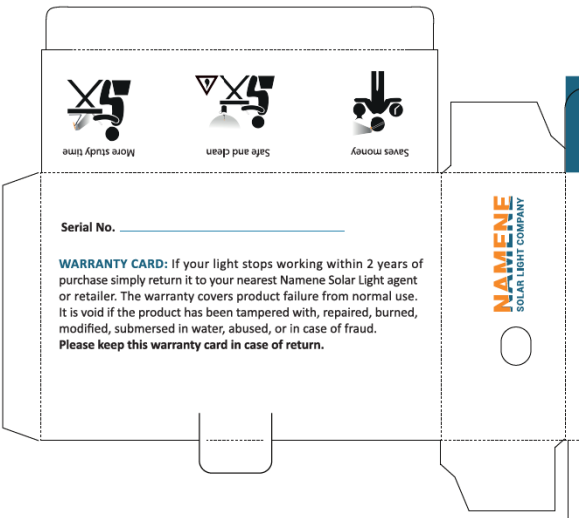
B.2. Applicability of methodology (ies)

The table below presents the applicability justification of the methodology AMS-III.A.R v7.0 for the distribution of solar lights.

Table 3 – Applicability justification for selected methodology

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
Paragraph 3. This methodology is applicable only to project lamps whose batteries are charged using the following options: (a) Charged by a renewable energy system included as part of the project lamp (e.g. a photovoltaic system or mechanical system such as a hand crank charger); (b) Charged by a standalone distributed generation system (e.g. a diesel generator set) or a mini-grid, i.e. that is not connected to a national or regional grid; (c) Charged by a grid that is connected to regional/national grid; (d) Charged by a combination of the methods described in options (a), (b) or (c) above.	a) The Project Lamps will be charged by renewable energy (a solar panel).
Paragraph 4. At a minimum, project lamps shall be certified by their manufacturer to have a rated average operational life of at least:	Yes. The Project Lamps will be certified. There is no certification available for an operational life of at least 10,000 hours, therefore, the Project Developer has chosen the shorter measurement period of 2,000 hours as per paragraph 19.b. of AMS-III.AR

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
(a) 5,000 hours for Option 1, paragraph 18; (b) 10,000 hours for Option 2, paragraph 19.	Version 7.0<i>"As an alternative to long-term measurement of light output over the full lifetime of the lamp, a shortened measurement period of 2,000 hours may be chosen. If a 2,000 hour test period is used, the relative luminous flux shall not decrease by more than 15 per cent during the 2,000 hours of continuous operation."</i> Lighting Global has certified that the SM200 solar light passes these criteria and therefore justifies the selection of option 4b. Further solar light models distributed by the project will comply with this requirement.
Paragraph 5. Rated average life is the life certified by the manufacturer or responsible vendor as being the time at period over which the lamp's initial light output will decline by no more than 30 per cent. In addition, for project lamps charged using the options from paragraphs 3(c) or 3(d), if a grid that is connected to regional/national grid is one of the sources used to charge the project lamps, the manufacturer shall certify that the battery-charging-circuit efficiency of the project lamps, at the time of the purchase, is at least 50 per cent. For project lamps charged under the options indicated in paragraph 3(b) or 3(d), if a mini-grid is one of the sources used to charge the project lamps and the mini-grid or distributed generation system is not entirely powered by renewable energy generation unit(s), the manufacturer shall certify that the project lamp's battery charging circuit efficiency, at	Under paragraph 19.b. of the methodology an alternative to Paragraph 5 is available and is highlighted in bold below..."<i>At a minimum, project lamps must be certified by their manufacturer to have a useful operational life of 10,000 hours. Within this time span, the relative luminous flux shall not decrease by more than 30 per cent as per equation (1). Such claims shall be confirmed by a third-party testing organization using an applicable standard and testing protocol. As an alternative to long-term measurement of light output over the full lifetime of the lamp, a shortened measurement period of 2,000 hours may be chosen. If a 2,000 hour test period is used, the relative luminous flux shall not decrease by more than 15 per cent during the 2,000 hours of continuous operation."</i> The Project solar lights will be certified to meet these criteria.

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
the time of purchase, is at least 50 per cent. Paragraph 6. The Project lamp will meet warranty requirements of Lighting Global Minimum Quality Standard. The project lamps shall have a warranty of a minimum one year from the time the end-user takes ownership or begins using the lamp. At a minimum, the warranty shall cover free replacement or repair of any failed lamps, batteries, and where applicable solar panels. The warranty shall be clearly communicated and supported through the supply chain and available to end-users of the project lamps during the warranty period. In a situation where the project lamps are distributed through intermediaries, the one year warranty shall commence from the time that the project lamps are distributed to end-users. The full warranty terms shall be available in writing, in a regionally appropriate language and included with each unit.	Yes. The Project Lamps shall be certified by Lighting Global Minimum Quality Standard (now VeraSol) and meet its warranty requirements. The project lamps or SHS shall have a minimum warranty period of 1 year and will cover free replacement or repair of any failed lamps, batteries, and solar panels and the VPA Implementer shall clearly communicate the warranty terms to the end user and provide the support during the warranty period. For the end user, warranty terms are printed on the packaging of the solar light. An example of the warranty card is presented below:  For the distribution channel the warranty terms are included in the Standard Terms and Conditions of Sale and are extended to 30 months to allow for inventory ageing.
Paragraph 7. Project lamps shall meet or exceed the following minimum performance characteristics, which should be proven by third-party test results: Light Output – luminous flux of 25 lumens or illuminance of 50 lux over	Yes, the Project Lamps will have a minimum light output of 25 lumens or illuminance of 50 lux over an area greater than or equal to 0.1m² when suspended at a distance of 0.75 meters or self-supported and the light output shall not decline more

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
an area ≥ 0.1 m² when suspended at a distance of 0.75 meters or self-supported. The light output over a 2,000 hour lumen maintenance test should not decline by more than 15%;	than 15% over a 2,000 hour luminance test. Furthermore, solar lights will be certified by VeraSol under IEC/TS 62257-9-5:
Paragraph 7b. Run Time and Battery Capacity - Daily Burn Time (DBT) shall meet the following requirements: (i) DBT shall be equal to or greater than 4 hours; (ii) For charging Option 3(a) with solar PV, the DBT is defined by the Solar Run Time for the project lamp (as determined per paragraph 9(g)); (iii) For other technologies in Option 3(a), the DBT is defined based on typical expected patterns of use; (iv) For charging Options 3(b) and 3(c): a. The maximum claimed DBT shall be less than or equal to the typical capabilities of the regional or local energy system at delivering reliable power sufficient for recharging; b. The autonomous (full battery) run-time of the project lamps shall be equal to or greater than 200 per cent of the DBT of the project lamps; c. The project lamp shall be fully recharged from a discharged state after eight hours of charging.	The Project Lamps will have a SRT greater than 4 hours at IEC standard 5 KWh per m².
Paragraph 8a. The project design document shall explain the proposed distribution method of the project lamps. It shall also explain how the proposed project activity shall ensure that the replaced baseline lamps are those that directly consume fossil fuel. This can be done through documentation of the	The target group for deployment of project solar lights is rural households living in communities that are located outside the boundaries or limits of the cities and towns in Zimbabwe. Target users will have no access to grid electricity and will be extensively using fossil fuel-based lighting as a default. The method of distribution is described in section A.3. of the VPA-DD.

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
common practice of fuel usage for lighting in the project region (e.g. based on representative sample surveys, official data or peer reviewed literature) that demonstrates that fossil fuel is a commonly used fuel for lighting;	Official reports were used to determine the electrification rate in the Project Area and a baseline survey was conducted to determine the usage and expenditure in fossil fuels for lighting. The method and results from the sample survey are presented in section B.4 of this document. Furthermore, agents of the project will ask potential end users of the Project Lamps which forms of lighting they currently use at home as part of an onboarding questionnaire, the results of which will be stored on the project database.
Paragraph 8b. Encourage the consumers, targeted by the project activity, to use the project lamps and discourage hoarding.	The VPA implementer will be selling the Project Lamps at a minimum price, to attach a value to the lights to encourage usage. Given that there are no expenditures attached with the use of the Project Lamp and it leads to savings on lighting expenditure, users will be encouraged to use it as their source of lighting. Furthermore, the VPA implementer will limit the number of Project Lamps to a single customer to avoid hoarding. Furthermore, the warranty will be linked to the personal details of the buyer, therefore incentivising them to keep the light to ensure the warranty remains valid. Finally, the VPA implementer will conduct from time to time awareness campaigns among users to promote the correct use and recharging of the Project Lamps and deal with any customer service, as required.
Paragraph 8c. Eliminate potential double counting of emissions reductions.	Project Lamps will be distinctively marked to be recognised as part of a unique VPA and will also include a unique serial number to avoid potential double counting of emission reductions. End users are informed of their carbon rights waiver when taking part in the VPA during briefings prior to the subsidised sale of Project solar lights. A Project Instruction

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
	and Process Manual will be used by Local Distribution Partner(s) with explicit instructions and wording to be used. This avoids the risk of double counting by end users. For Local Distribution Partner(s), a clause is included in a standardised Distribution Agreement to ensure full and uncontested legal ownership of carbon credits generated during the crediting periods and avoids the risk of double counting.
Paragraph 8d. Ensure compliance with prevailing regulations pertaining to use and disposal of batteries.	To date, Zimbabwe has no legislation or policy on electronic waste management. The available Environmental Waste Management Act (20:27)⁷ only prohibits the discharge of hazardous substances into the environment, but there is no specific legislation regulating batteries. Project Lamps will have easily field replaceable batteries and the VPA will have a procedure to collect end of life or failed batteries by field sales agents. Representatives will be trained to diagnose faults and replace batteries in the field. When lights need to be fixed they will be replaced and there will be repair centres in different parts of the country for basic repairs, and lights which can't be fixed there will be shipped for refurbishment or disposal. Collected batteries will be safely stored and appropriately disposed or recycled as per the country regulations and existing facilities. An example of such a light is presented below:

⁷ [Environmental Management Act \[Chapter 20:27\]. | UNEP Law and Environment Assistance Platform](#)

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
	
Paragraph 9. The VPA Design Document shall include the minimum requirements for the design specifications of project lamps.	The minimum specifications of the project lamps for this VPA are presented in Table 4. Other lights that meet all the minimum PoA and methodology criteria might be added in the future.
Paragraph 10. Measures are limited to those that result in emissions reductions of less than or equal to 60kt CO₂e equivalent annually.	The VPA will restrict deployment to up to 652,000 units of solar lights (each light reducing 0,092 tCO₂e/year) which would result in annual emission reductions under the 60,000tCO₂e threshold.
Paragraph 38. If monitoring is required, per paragraphs 34–37, to determine the percentage of project lamps distributed to end-users that are operating and in service, such monitoring will take place in the third year of the crediting period of each VPA and the results shall be used for operational years 4, 5, 6 and 7 of the project lamps of that VPA.	The VPA Implementer will conduct ex-post monitoring surveys to determine the percentage of project lamps distributed to end users that are operating and in service during the third year of the crediting period of the VPA. The results of the monitoring in year 3 may be used for operational years 4, 5, 6 and 7 of the project lamps of this VPA.
Paragraph 39. For Option 2, paragraph 19, project lamps shall be marked for clear, unique identification to associate them with each unique VPA.	Project Lamps will be distinctively marked to recognised as part of this VPA or PoA and will also include a unique serial number as indicated in Paragraph 8 above and the eligibility criteria for inclusion in the VPA.
Paragraph 40. The option in paragraph 36 which allows the use of the results of the sampling surveys of the first batch as a proxy to	The VPA Implementer will only apply the results of the sampling surveys of the first batch as a proxy to subsequent batches

CRITERIA – AMS-III.AR VERSION 7.0	CONDITION RELATED TO SOLAR LIGHTS
subsequent batches should only be applied to the same CPA to which the first batch belongs.	within the same VPA to which the first batch belongs.

Table 4 - Project Lamps minimum technical specifications:

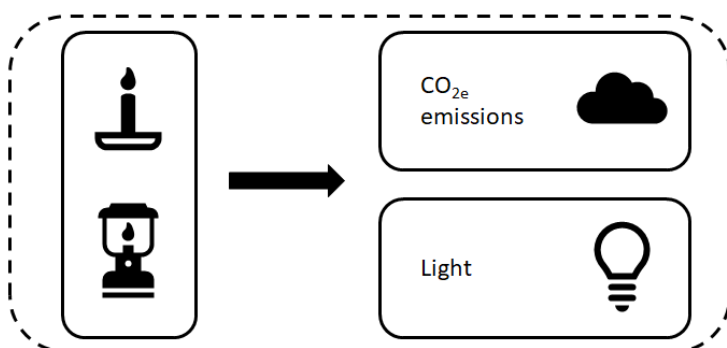
Specification	Values
(a) Lamp wattage (Watts)	0.3
(b) Luminous flux output (lumens) High or Medium setting	25
(c) The type and rated capacity of the renewable energy equipment used for battery-charging (in Watts);	Solar photovoltaic cells (PV) 0.28
(d) Battery type	Lithium-iron-phosphate battery (LiFePo4)
(d) Battery nominal voltage (V)	3.2
(d) Battery Capacity (mAh)	500
(d) Battery-charging circuit efficiency (%)	90
(e) Type of charge controller	Deep discharge and overcharge protection
(f) Autonomous Run Time - AT (hours)	5
(f) Daily Burn Time -DBT (hours)	5 (high setting) / 12 (low setting)
(g) Solar Run Times – SRT (hours)	5 (high setting) / 12 (low setting)
(h) Where applicable, the amount of time to fully charge the product using mechanical means or a centralized charging system	Not applicable
(i) Physical protection against environmental factors	IP44 (protection against dust and water ingress)

Certification	VeraSol ⁸
Rated lamp life (Hours);	10,000

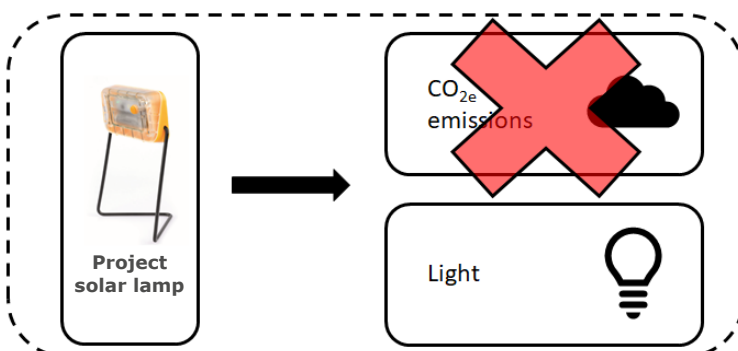
B.3. Project boundary

The project boundary includes the project lamps as well as the charging systems. In the project scenario the project lamps are charged by a renewable energy system (solar PV) therefore the project boundary includes the physical, geographical site of the renewable energy system, which is the location where the project lamps will be distributed. The project boundary in the baseline as well as in the project scenario is presented below:

Project boundary baseline



Project scenario



Source	GHGs	Included?	Justification/Explanation
Baseline scenario Combustion of kerosene for lighting	CO ₂	Yes	Major source of emissions from kerosene lamps

⁸ [VeraSol Product Database](#)

	Combustion of candles for lighting	CO ₂	Yes	Major source of emissions from paraffin candles
Project scenario	None	CO ₂	NO	No emissions from rechargeable solar lights as per Methodology AMS-III.AR Version 7.0 Section 5.4 - Project Emissions (Paragraph 29).

B.4. Establishment and description of baseline scenario

Introduction

The use of kerosene and other fossil fuels for lighting by target users is the applicable baseline scenario. As per the methodology of AMS-III.AR Version 7.0, in the absence of the project activity, the baseline scenario would be the projected use (combustion) of fossil fuels to meet similar lighting needs as those provided by the Project Lamps. In the absence of the project activity, rural households in the provinces targeted by the VPA predominantly use kerosene lamps and/or paraffin candles for lighting.

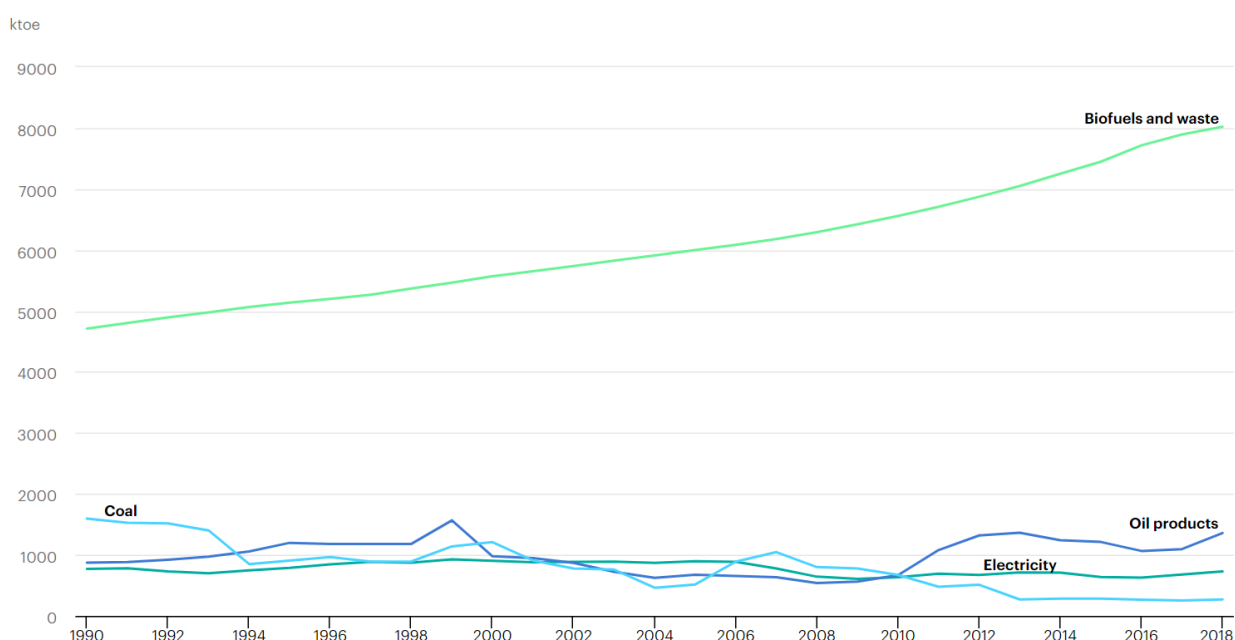
Regarding the baseline scenario for GHG emissions, the Project is using the CDM methodology AMS-III.AR Version 7.0. This methodology provides for a default annual baseline emissions factor for the project lamps. As per the methodology, there are no project emissions (PE_y = 0) if the project lamp charging mechanism utilized is from a renewable energy, which in this case, all project lamps will be charged by the integrated solar panel.

According to the World Bank, as of 2019 only 41% of the population of Zimbabwe holds access to electricity. This is representing a steady increase of 1% over the last two years however overall is a fall from its height at 2012 of 44%. In rural areas, the electrification rate is 20%, again representing a steady increase of 1% in the past couple years but a drop from its height in 2012 of 23.8%⁹.

⁹<https://data.worldbank.org/indicator/EG.ELC.ACCS.RU.ZS?locations=ZW>

The United Nations Household Size and Composition Around the World 2017 states that Zimbabwe has an average household size of 4.1¹⁰. According to USAID, there are 9.5 million citizens living in rural areas in the country. This means that there are approximately 2,317,000 households in rural Zimbabwe, and 1,853,700 households without access to electricity¹¹. The Total Final Consumption (TFC) shows that the population of Zimbabwe predominately use biofuels and waste as their primary source of energy¹². Specific statistics on the household energy mix are unavailable, as well as recent statistics on rural income.

Total final consumption (TFC) by source, Zimbabwe 1990-2018



Although more recent statistics on household energy expenditure are unavailable, a survey in Gutu district from 2015 states that the typical household expenditure on kerosene and candles is between \$8 and \$15 p/month, meaning an annual

¹⁰https://www.un.org/en/development/desa/population/publications/pdf/ageing/household_size_and_composition_around_the_world_2017_data_booklet.pdf

¹¹<https://www.usaid.gov/zimbabwe>

¹²[Data & Statistics - IEA](#)

expenditure of between \$100 and \$200. The report notes that, this expenditure was “still not enough to enable children to read at night, for farmers to extend their working hours or for women to give birth safely.”¹³

The National Renewable Energy Policy of 2019¹⁴ is the most recent policy paper released by the government of Zimbabwe. In this paper, the government states it will work alongside the rural electrification fund to increase rural electricity access by

- The renewables based off-grid technologies which can be used for electrification are mini grid projects and others. Standalone decentralised systems which can be used are solar lighting systems, solar home systems, solar lanterns, and biogas electrification plants.
- Special provision for tariff shall be made to make it affordable while ensuring a desired return to investors.
- The programme shall promote local community participation in the development and operations of the project.
- It shall also support the development of independent grids supplied by micro-hydro and pico-hydro, biomass gasifiers and solar PV systems in dispersed distant settlements.
- The programme shall prioritise supporting electrification for productive uses and key social services. Captive generation shall be promoted through incentives.
- The programme shall promote public private partnership and private participation through tendering.

A load forecast of domestic households and institutions in rural off-grid areas in the Rural Energy Master Plan indicate that the potential additional rural electricity demand by the year 2034 is between five hundred megawatts (500 MW) and nine hundred megawatts (900 MW), under the ambitious assumption that all rural customers will be grid-connected. It is noted in the there are several off-grid technologies and other clean energy solutions that can improve energy access in the distant rural areas of

¹³<https://www.scribd.com/document/340882047/Transforming-Lives-in-Zimbabwe-Rural-Sustainable-Energy-Development-Project>

¹⁴<https://www.techzim.co.zw/2019/08/download-ministry-of-energy-releases-national-renewable-energy-policy/>

Zimbabwe like mini grid, micro grid, solar lighting, solar home systems, small hydro, solar water heaters, clean cook stove and RE based water pumps and systems.

The Project Developer has expanded this assessment with a baseline survey that was undertaken in May 2021. The planning and execution of the survey was based on CDM-EB67-A06-GUID¹⁵ (Guidelines - Sampling and surveys for CDM project activities and programmes of activities) (hereinafter "The Guidelines") and CDM-EB50-A30-STAN¹⁶ (Standard - Sampling and surveys for CDM project activities and programmes of activities – Version 8.0 – November 2019) (hereinafter "The Standard").

The sampling size was determined by minimum 90 per cent confidence interval and the 10 per cent maximum error margin and the size of the sample shall be 246 meeting the requirement of not being less than 100. The survey has a random distribution and is representative of the target population, more so sampling was statistically robust and relevant. Data collection and capturing was completed using the Namene Solar App.

The objective of the survey was to determine the typical kerosene or candles expenditure and consumption patterns for lighting in households in Zimbabwe more so as per District level, particularly in rural areas that have no access to electricity and clean lighting sources. The baseline survey also produced an outcome on how much impact the project will have about other social and environmental impacts, such as the cost benefit analysis on households by using solar for lighting instead of refuelling kerosene lamps or buying candles.

Target Population

The population mainly targeted in this survey was the rural households who are not connected to the grid. The districts selected were the ones where the project developer has access to the local leadership. They also represent the whole population of Zimbabwe since they are from all the 8 rural provinces, therefore giving a better representation in terms of monthly incomes and age demographics. These are also

¹⁵ [meth_guid46.pdf \(unfccc.int\)](#)

¹⁶ [CDM Catalogue of Decisions: Standard. Sampling and surveys for CDM project activities and programme of activities. \(version 04.0\). \(unfccc.int\)](#)

the areas with the lowest grid connectivity as of the ZIMSTAT statistics of the last current physical census in 2012.

Sampling frame

In this case, the sampling frame was the whole population that was indicated as the target population thus all rural off-grid households (rural households consist of 68% of the total population in Zimbabwe and urban 32%) that do not have solar systems in the indicated Districts. Given the sampling frame, it was not practical for to have the information and interview them all so the survey had a sampling frame of 124,4341 households that was randomly selected to represent the whole total population by the assigned local leaders in different districts. This was done to have meaningful representation and randomized approach for selecting the sampling population.

Sampling method

For this survey, the project developer selected the stratified random sampling method as the most effective and efficient method. Since the sampling frame was be classified into subgroups of unequal proportions, it was effective to make use of stratified. Some of the factors that led to choose of this method over the other sampling methods are:

- It accurately reflects the population in the survey since we will stratify the entire population before applying random sampling methods.
- Heterogeneity of the selected Districts due to differences in economic backgrounds.
- It provides better coverage of the population; we will have control over the districts to ensure all of them are represented in the sampling.
- Differences in climatic Regions and literacy rate in the districts.
- Levels of spending, for example those who live in areas with a lot of minerals spends more.
- It will also guard against the unrepresented sample.
- It provides greater precision and requires a smaller sample, which will help in cost cutting given the pandemic crisis.

Sample Size

With the stratified sampling, the sample will be split into sub-groups so we will have sub samples from different strata. The sample size to be taken in this survey is from a finite population of 124,434¹⁷. The level of confidence\precision to be implemented will be 90/10. Some of the parameter estimates we will use are the mean and standard deviation. Since the survey has been categorized as a small scale, we will use the 90% confidence that the margin of error in our estimate is not more than $\pm 10\%$. We will conduct survey on the solar lights in 25 different districts. We will mainly make use of previously calculated Estimates of proportion of households with access to clean lighting thus the ones on grid, ones who use solar systems, torches, and solar lamps in that specific region. These proportions of households are taken from the previous surveys we have conducted in conjunction with information from the local leader's database.

Sample size calculation

Sampling Method : Stratified Sampling

Calculation Method : Precision via Confidence Interval Based on Proportion Allocation

Confidence/Precision : 90/10

Input	Value	Source / comments
Confidence level	90%	Default from Methodology
Precision	10%	Default from Methodology
z-multiplier	1.645	Default from Methodology
Overall proportion	0.512	Calculation
Overall variance	0.229	Calculation
V, ratio of variance to proportion squared	0.873	Calculation
Population size, N	124,434	Census
Total Number of Strata	25	25 districts in Zimbabwe
Predicted sample size, n	236	Calculation
Final Sample Size	246	Sum from table below

District	Expected Proportion	Total Households ¹⁹	Proportional Sample Allocation
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¹⁷ Number of households from the Districts to be covered as per the Attached ZIMSTAT 2012 Census Report

¹⁹ This is the total off grid rural households as per 2012 census

	with Lighting ¹⁸		
Bikita	0.34	4,425	9
Bindura	0.47	4,007	8
Buhera	0.41	6,167	12
Chikomba	0.44	3,407	7
Chimanimani	0.34	3,252	7
Chipinge	0.38	8,004	16
Chiredzi	0.47	7,607	15
Chirumanzu	0.27	1,988	4
Chivi	0.46	4,358	9
Goromonzi	0.63	5,029	10
Gutu	0.33	5,590	11
Gweru	0.67	6,282	12
Insiza	0.34	2,435	5
Makoni	0.69	7,689	15
Masvingo	0.68	7,482	15
Mberengwa	0.35	5,184	10
Mudzi	0.42	3,683	7
Murehwa	0.67	4,590	9
Mutare	0.71	10,995	21
Mutasa	0.68	4,725	9
Mwenezi	0.42	3,547	7
Nyanga	0.47	3,368	7
Shurugwi	0.37	2,496	5
Zaka	0.41	5,215	10
Zvishavane	0.69	2,909	6
Total		124,434	246

Population (N_i) – This is the total population as per districts, its total will give us the sampling frame (124 434).

Expected Proportion with Lighting (p_i) – This is the proportion of households who are on grid or who use solar systems as per records from Local Leaders heads.

Proportional Sample Allocation (n_i) – This is the sample size per district basis, if added they will sum up to our survey sample size.

Some key Elements:

- $N = 124\,434$

¹⁸ These proportions are from the local leaders' database and previously undertaken surveys

- Confidence level – 90%
- Precision – 10%
- Z-value – 1.645
- We let our random variable to follow a normal distribution.

Sample Size Calculations:

The Equation for total sample size (n) calculation:

$$n \geq (1.6448)^2 * (N) * (V) [(0.1)^2 * (N-1)] + (0.6448)^2 * V$$

Where $V = SD^2\alpha$, SD^2 is the overall Variance and α is the overall proportion.

In calculating the number of households in the sample that come from each district we used the equation:

$$n_i = (Sf_i) * (n) / N \text{ whereby } Sf_i \text{ is the total number to be surveyed per district and } i=1, 2, 3, \dots, 25$$

Taking an assumption that the proportions of individuals from the different districts in the sample are the same as the proportions in the population.

The overall variance (SD^2) was calculated as:

$$\frac{1}{N} \sum_{i=1}^{25} Sf_i (p_i - \alpha)^2 \text{ where by } (p) \text{ is the Response Proportion from previous surveys and } (\alpha) \text{ the overall proportion as: } \frac{1}{N} \sum_{i=1}^{25} [Sf_i * (p_i)]$$

Application:

$$\begin{aligned} SD^2 &= \frac{1}{1124\,434} [4\,425(0.34-0.342)^2 + 4\,007(0.47-0.472)^2 + 4\,167(0.41-0.42)^2 + \dots \\ &\quad + 2\,909(0.69-0.692)^2] \\ &= \frac{63\,719}{1124\,434} \\ &= 0.512 \end{aligned}$$

$$\begin{aligned} \alpha &= \frac{1}{1124\,434} [4\,425(0.34) + 4\,007(0.47) + 4\,167(0.41) + \dots + 2\,909(0.69)] \\ &= \frac{28\,475}{1124\,434} \\ &= 0.2288 \end{aligned}$$

Now we have the overall variance (SD^2) and the overall proportion (α) so we can now calculate:

$$\begin{aligned} V &= SD^2\alpha^2 \\ &= 0.23(0.512)^2 \\ &= 0.873 \end{aligned}$$

Substituting in V into our sample size equation $\{n \geq (1.6448)^2 \cdot (N) \cdot (V) [(0.1)^2 \cdot (N-1)] + (0.6448)^2 \cdot V \}$ gives:

$$\begin{aligned} n &\geq (1.6448)^2 \cdot (124\,434) \cdot (0.873) [(0.1)^2 \cdot (124\,434-1)] + (0.6448)^2 \cdot 0.873 \\ &= 293\,8541\,246.7 \\ &= 235.7071 \end{aligned}$$

So we obtained that $n \geq 235.7071$ and rounding it off to the nearest whole number we get 236 thus the minimum sample size we can consider for the survey. This then needs to be divided up according to the size of each district to get the number of households that should be sampled in each district.

$$n_i = (Sfi) \cdot (n)N$$

$$n_1 = 4\,425 / 124\,434 \times 236 = 9$$

$$n_2 = 4\,007 / 124\,434 \times 236 = 8$$

$$n_3 = 6\,167 / 124\,434 \times 236 = 12$$

$$n_{24} = 5\,215 / 5\,760 \times 236 = 10$$

$$n_{25} = 2\,909 / 124\,434 \times 236 = 6$$

Kindly note all are rounded to the nearest whole number so as to represent number of households.

The results of the survey were that out of the 246 individuals interviewed, 146 used fossil fuel forms of lighting, hence representing the most common source of lighting. These being a kerosene lamp (90), paraffin candle (124) or combination of the two. This means that 59% of the surveyed population used fossil fuel forms of lighting. Other sources of lighting representing the rest of the sample (100), including battery powered torches, firewood, solar lights and animal dung. The results and calculations of the baseline scenario for SDG 1 are presented in section B.6.1. Therefore, a significant amount of the population consumed a combination of kerosene and candles to enable a small-scale project in line with CDM methodology AMS-III.AR Version 7.0.

B.5. Demonstration of additionality

The Gold Standard Community Activity Requirements v1.2 in Paragraph 4.1.9 states that:

Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification:

(a) Positive list (Annex B of this document)

(b) Projects located in LDC, SIDS, LLDC

(c) Microscale projects

The table below is only applicable if the proposed project is deemed additional, as defined by the applied approved methodology or activity requirement or product requirement.

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Paragraph 4.1.9 of the GS Community Services Activity Requirements (Version 1.2): "Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: (a) Positive list (Annex B of this document) (b) Projects located in LDC, SIDS, LLDC (c) Microscale projects"
Describe how the proposed project meets the criteria for deemed additionality.	This VPA is located in Zimbabwe, which is a Landlocked Developed Country (LLDC) as defined by the United Nations²⁰ and therefore can be considered as deemed additional as per the paragraph 4.1.9 (b) above and consequently the Financial Additionality is not required to be proved at the time of Design Certification.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

SDG Impact

²⁰ [List of Landlocked Developing Countries | UNCTAD](#)

Sustainable Development Goals Targeted	Most relevant SDG Target	Indicator (Proposed or SDG Indicator)
SDG 1 – No Poverty	Target 1.1- By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day.	Average household savings i.e., decrease in expenditure on lighting
SDG 3 – Good Health and Wellbeing	Target 3.9 - By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	Number of households visited medical facilities/dispensary for treatment of respiratory issues etc. such as cough, shortness in breath, pneumonia and other respiratory issues
SDG 4 – Quality Education	Target 4.4 - By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship	Number of employees provided skill development training
SDG 7 – Affordable and Clean Energy	Target 7.1: By 2030, ensure universal access to affordable, reliable and modern energy services.	Number of beneficiaries: Households with access to clean and affordable solar lights
SDG 8 – Decent Work and Economic Growth	Target 8.5 - By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Total number jobs created as a result of the project
SDG 13 – Climate Action	Target 13.3 – Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning.	GHG Emission reductions

The households involved in the project, they use traditionally fossil fuels for lighting, such as kerosene and candles, which result expensive, inefficient and polluting to the environment. Through the implementation of the project by introducing solar lights, the avoidance of fossil fuels will result in GHG emission reduction (contributing to SDG 13). The introduction of solar lights will improve the quality of illumination in households and safety on the roads, due to the portability of the lights (contributing to SDG 7). The use of the solar light will result in continued savings from expenditures in expensive kerosene, candles or batteries, hence contributing to reduce poverty (SDG 1). By avoiding the consumption of kerosene for lighting, the project also reduces black carbon emissions from the combustion of kerosene and candles, hence improving indoor air quality (SDG 3). The project will contribute to SDG 4 (Quality of Education) by increasing the technical and commercial skills of people employed by the project through regular training. It also contributes to the creation of new job opportunities in a promising growing industry (renewable energy), therefore contributing to SDG 8 (Decent Work and Economic Growth).

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

For SDG 1 – No Poverty:

The Project will contribute to SDG1 due to the financial savings to the project beneficiaries from the avoidance of purchasing kerosene, candles or disposable batteries for lighting by owning a project solar light. The Project Proponent developed its own SDG 1 impact quantification method by comparing the expenditures in lighting during the baseline scenario, compared to the expenditures during the project scenario (when end users own a Project solar light).

The selected impact indicator for SDG 1 will be financial savings by the participant households in the project during the crediting period. This quantification method is described as follows:

Applied methodology/approach	Equation/calculation
Target 1.1- By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day.	Where:

Approach: Monitor the money savings in households due to reduced fuel consumption by using Project Lamps	The project outcomes in terms of financial savings per household on year y is given by the following equation: $Ex_{sl,y} = (Ex_{h,b} - Ex_{h,p}) \times OF_y \quad \text{Equation (1)}$ Where: $Ex_{sl,y}$ = financial savings per household on year y $Ex_{h,b}$ = baseline expenditure per year per household on lighting $Ex_{h,p}$ = project expenditure per year per household on lighting OF_y = Percentage of project lamps distributed to end users that are operating and in service in year y. Assumed to be equal to 100 per cent for years 1, 2 and 3, and equal to the value determined as per the sample survey described in section B.7.2, for years 4, 5, 6 and 7. Ex-ante assumed to be 80% for years 4, 5, 6 and 7.
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There has not been any distribution of Project Lamps in this VPA. The Project scenario will be determined using a sample survey (ex-post, one-off, during the project's first monitoring period) to quantify the expenditure of fossil fuels and batteries for lighting after the households received a Project Lamps. These expenditure values will be then used for the calculation of economic savings due to reduced expenditure in lighting by using the solar light during the rest of the crediting period.

For SDG 3 – Good Health and Well Being

The project will reduce Reduction of household air pollution by decreasing household black carbon emissions. Indoor air quality is improved when lighting from kerosene lamps is replaced with solar lighting.

Applied methodology/approach	Equation/calculation
Target 3.9 - By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination. Indicator: Number of households visited medical facilities/dispensary for treatment of respiratory issues etc. such as cough, shortness in breath, pneumonia and other respiratory issues Approach: Monitor the % of customers who are experiencing better quality indoor air and have reduced incidences of respiratory or eye issues since deployment of the solar lights.	Follow-up surveys to end users to include questions surrounding air quality and health related to respiratory and eye issues. Results delivered as a percentage of households who have stated that they experienced less incidences of respiratory or eye issues following deployment of solar lights.

For SDG 4 – Quality Education

The project aims to hire people from the local communities to perform vital roles in distribution of the Project Lamps, such as in sales and using the monitoring system as well as technical after sales service. The project developer aims to provide vocational and technical training sessions to these employees in order to increase their skills that can be used even when the project is complete. These training sessions will be recorded with a full list of all attendees.

Applied methodology/approach	Equation/calculation
Target 4.4 - By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship	To record number of training sessions ran for employees, including attendance lists

Approach: Record the number of employees receiving skill based training.	
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For SDG 7 – Affordable and Clean Energy

This VPA will contribute to SDG 7 by increasing the population accessing clean and affordable solar lights. VPA outcomes will be based on the total number, n , of solar lights deployed under this VPA.

Applied methodology/approach	Equation/calculation
Target 7.1: By 2030, ensure universal access to affordable, reliable and modern energy services. Indicator; Number of beneficiaries: households. Approach: Monitor the number of Project Lamps distributed under the project and by the number of Project Lamps Distributed per household.	Project Lamps distribution records. $\text{Net benefit SDG 7} = \text{Additional households accessing clean source of lighting} = N \text{ of project lights distributed} / \text{number of lights distributed per household} .$

For SDG 8 – Decent Work and Economic Growth

Namene Solar creates employment opportunities in each country where it operates, in sales, monitoring or after-sales services, as well as management. These employment opportunities are tracked.

Applied methodology/approach	Equation/calculation
Target 8.5 - By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with	Employment records $\text{Net Benefit SDG 8} = \text{Employment and income generation (Number of person (male and female) hired by the Project)}$

disabilities, and equal pay for work of equal value Approach: Recording the number of contracted employees (male / female) in the project	
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For SDG 13 – Climate Action

The methodology selected AMS-III.AR v.7.0 provides for a default value annual baseline emissions factor for the project lamps. The following assumptions are made about the equivalent baseline lighting system:

$$DV = FUR \times O \times U \times EF \div 1000 \times LF \times n \times NTG \quad \text{Equation (2)}$$

Where:

DV = Lamp Emission Factor (default is 0.092 tonnes CO_{2e} per project lamp)

FUR = Fuel use rate (0.03 litres/hour)

O = Utilization rate (3.5 hours/day)

U = Annual utilization (365 days/year)

EF = Fuel emissions factor (2.4 kgCO₂/litre)

LF = Leakage factor (1.0)

n = Number of fuel-based lamps replaced per project lamp (1.0)

NTG = Net-to-gross adjustment factor (1.0)

For the purposes of this Project, the project developer assumes that DV is set at the default defined in Equation (2) per Project Lamps.

Baseline emissions per Project Lamps are calculated per equation (3) of the selected methodology AMS-III.AR v.7.0:

$$BE_y = DV \times GF_y \times DB_y \quad \text{Equation (3)}$$

Where:

BE_y = Baseline emissions per project lamp in year y (tonnes CO_{2e})

DV = Lamp Emission Factor (default is 0.092 tonnes CO_{2e} per Project Lamps)

GF_y = Grid Factor in year y , Equal to 1.0 when charging option defined in paragraph 3(a) is used; Equal to 1.0 if the project activity is for off-grid households/communities (defined as no grid access or less than 12 hours grid availability per day on an annual average basis); Otherwise, it is equal to 1.0 minus (the fraction of time grid is available to the target households and communities/users in the region of project activity)

DB_y = Dynamic Baseline Factor (change in baseline fuel, fuel use rate, and/or utilization during crediting period) in year y . Calculated as either:

Option 1: default of 1.0 in the absence of relevant information.

Option 2: value of $1.0 + FF_g$ where FF_g is the documented national growth rate of kerosene fuel use in lighting from the preceding years (use the most recent available data for three- or five-years average (fraction))

The project developer has selected the following values for these parameters based on the options available under the selected methodology:

DV Default value of 0.092 tonnes CO_{2e} per project light.

GF_y Equal to 1.0 as charging option defined in paragraph 3(a) is applicable to the Project solar light.

DB_y Option 1: default of 1.0 as relevant information on the national growth rate of kerosene fuel used in lighting from the preceding years is not available.

The Project emissions from the distribution of solar lights, PE_y , are zero as the charging option defined in paragraph 3(a) is applicable to the Project solar light.

Given the above, the annual emissions reductions are calculated as:

Annual emission reductions are calculated as:

$$ER_y = \sum N_{i,j} \times (BE_{y,i} - PE_{y,i,j}) \times OF_{y,i,j} \quad \text{Equation (4)}$$

Where:

ER_y = Emission reductions in year y (tonnes CO_{2e})

$N_{i,j}$ = Number of project lamps distributed to end users of type i with charging method j

$OF_{y,i,j}$ = Percentage of project lamps distributed to end users that are operating and in service in year y , for each lamp type i and charging method j . Assumed to be equal to 100 per cent for years 1, 2 and 3, and equal to the value determined in paragraph 30, for years 4, 5, 6 and 7. Ex-ante assumed to be 80% for years 4, 5, 6 and 7.

Under AMS-III.AR v.7.0 Section 5.5 paragraph 33; the emissions reductions shall be considered from the date of distribution of the project solar light to end users.

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter	DV
Unit	Tonnes CO ₂ e per year per Project Lamp
Description	Lamp Emission Factor
Source of data	AMS-III.AR V.7.0
Value(s) applied	0.092
Choice of data or Measurement methods and procedures	Default value per project lamp as per the methodology AMS-III.AR V7.0
Purpose of data	Supports calculation of Baseline Emissions per project lamp per year (tonnes CO ₂ e), Bey

Additional comment	
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SDG 1.1.1

Data/parameter	$Ex_{h,b}$
Unit	\$ USD
Description	Annual expenditure on lighting per household in the baseline scenario
Source of data	Baseline assessment study by Project Proponent in Zimbabwe 2021
Value(s) applied	303.93
Choice of data or Measurement methods and procedures	The data from a baseline assessment study in the country is used as it shows the spending expenditure on lighting per household on fossil fuel methods of lighting
Purpose of data	Supports calculation of financial savings per solar light due to avoided purchases of non-renewable energy sources for lighting.
Additional comment	

B.6.3. Ex ante estimation of SDG Impact

The values and parameters used for the ex-ante calculation of baseline and project outcomes is summarised in the table below:

Table 6. Parameters and values used for the ex-ante estimation of SDG impacts

SDG	PARAMETER	VALUE	UNITS	SOURCE
SDG 1	Annual expenditure in lighting per household in the baseline scenario (Ex,b)	303.93	USD	Baseline assessment study by Project Proponent in Zimbabwe 2021

SDG 1	Annual expenditure in lighting per household in the Project scenario (Ex,p)	99.18	USD	70% reduction compared to the baseline scenario. Assumption based on Project Developer's experience. Actual data used for SDG calculation during the crediting period will be based on ex-post monitoring
SDG 3	% of customers who are experiencing better quality indoor air and have reduced incidences of respiratory or eye issues since deployment of the solar lights.	80	%	80% reduction compared to the baseline scenario. Assumption based on Project Developer's experience. Actual data used for SDG calculation during the crediting period will be based on ex-post monitoring
SDG 4	Number of employees receiving skill based training	30	Employees trained/ year	Based on Project Developer's estimation on the training modules and employees working in the project. Actual data used for SDG calculation during the crediting period will be based on ex-post monitoring
SDG 8	Number of contracted employees in the project	30	Employees	Based on Project Developer's estimation recruitment and work positions required by the project. Actual data used for SDG calculation during the crediting period will be based on ex-post monitoring
SDG 13	Lamp emission factor (DV)	0.092	tCO ₂ e/year	CDM small-scale methodology AMS-III.AR version 7.0
SDG 13	FUR = Fuel use rate)	(0.03	litres/hour	CDM small-scale methodology AMS-III.AR version 7.0
SDG 13	O = Utilization rate	3.5	(hours/day)	CDM small-scale methodology AMS-III.AR version 7.0

SDG 13	U = Annual utilization	365	(days/year)	CDM small-scale methodology AMS-III.AR version 7.0
SDG 13	EF = Fuel emissions factor	2.4	kgCO ₂ /litre	CDM small-scale methodology AMS-III.AR version 7.0
SDG 13	LF = Leakage factor	1.0		CDM small-scale methodology AMS-III.AR version 7.0
SDG 1	Purchase price of solar lamp	4.00	USD	Project Developer's own estimate. Assumes that for every lamp distributed, the user would pay \$4.00 per light on the year of distribution and \$1.00 for battery replacement in year 4 of distribution. Assumes change of battery on year 4 for only 10% of the deployed volume. This is a conservative scenario as batteries are designed to last up to 7 years
SDG 1 / SDG7 / SDG 13	Percentage of project lamps distributed to end users that are operating and in service in year 1 to 3 (OF _{y,i,j})	100	%	CDM small-scale methodology AMS-III.AR version 7.0
SDG 1 / SDG7 / SDG 13	Percentage of project lamps distributed to end users that are operating and in service in year 4 to 7 (OF _{y,i,j})	80	%	Project Developer's own estimate, based on fieldwork assessments

The table below presents the Project Developer assumptions used for the number of solar lights distributed during the first crediting period, based on the Project Developer's own estimates. The table also calculates the cumulative lights distributed as well as the cumulative lights in operation for a given year.

Table 7: Number of cumulative lights distributed and in operation during the crediting period

NUMBER OF SOLAR LIGHTS DISTRIBUTED		CREDIT PERIOD 1				
		Year 1	Year 2	Year 3	Year 4	Year 5
SOLAR LIGHTS DISTRIBUTED		100,000	250,000	302,000	0	0
CUMULATIVE LIGHTS DISTRIBUTED		100,000	350,000	652,000	652,000	652,000
OPERATING LIGHTS %	Year 1	100%	100%	100%	80%	80%
	Year 2		100%	100%	100%	80%
	Year 3			100%	100%	100%
	Year 4				100%	100%
	Year 5					100%
OPERATING LIGHTS (TOTAL)	Year 1	100,000	100,000	100,000	80,000	80,000
	Year 2		250,000	250,000	250,000	200,000
	Year 3			302,000	302,000	302,000
	Year 4				0	0
	Year 5					0
	Cumulative	100,000	350,000	652,000	632,000	582,000

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 1 – No Poverty

SDG 1.1 – Reduction of expenditures on fossil fuels for lighting in households, contributing to reducing extreme poverty and poverty

Per Household (2 Solar Light)	Baseline estimate (USD)	Project estimate (USD)	Net benefit (USD)
Year 1	303.93	99.18	204.75
Year 2	303.93	91.18	212.75
Year 2	303.93	91.18	212.75
Year 4	303.93	91.38	212.55
Year 5	303.93	91.18	212.75
Total	1,519.63	464.09	1,055.54
Total number of crediting years	5.00	5.00	5.00
Annual average over the crediting period	303.93	92.82	211.11

Overall Project	Baseline estimate (USD)	Project estimate (USD)	Net benefit (USD)
Year 1	15,196,260	4,958,878	10,237,382
Year 2	53,186,911	15,956,073	37,230,837
Year 2	99,079,616	29,723,885	69,355,731
Year 4	96,040,364	28,875,309	67,165,055
Year 5	88,442,234	26,532,670	61,909,564
Total	351,945,385	106,046,816	245,898,570
Total number of crediting years	5	5	5
Annual average over the crediting period	70,389,077	21,209,363	49,179,714

SDG 3

3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination

OVERALL PROJECT	% HH EXPERIENCING FEWER RESPIRATORY DISEASES AFTER ADOPTING SOLAR LIGHTING
Year 1	80%
Year 2	80%
Year 3	80%
Year 4	80%
Year 5	80%
Total number of crediting years	5
Annual average over crediting period	80%

SDG 4 – Quality Education

Target 4.4 - By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship

Number of employees delivered skills development training

Year	Baseline estimate (number of employees trained)	Project estimate (number of employees trained)	Net benefit (number of employees trained)
Year 1	0	15	30
Year 2	0	30	60
Year 2	0	30	60
Year 4	0	30	60
Year 5	0	30	60
Total	0	30	270
Total number of crediting years	5		

Annual average over the crediting period	0	27	54
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SDG 7 – Affordable and Clean Energy

SDG 7.1: By 2030, ensure universal access to affordable, reliable, and modern energy services.

Households benefiting from access to affordable and clean energy for lighting due to the project.

OVERALL PROJECT	HOUSEHOLDS BENEFITED
Year 1	50,000
Year 2	175,000
Year 3	326,000
Year 4	316,000
Year 5	291,000
Total number of crediting years	5
Annual average over crediting period	231,600

SDG 8: – Decent Work and Economic Growth

Year	Baseline estimate (number of jobs created)	Project estimate (number of jobs created)	Net benefit (number of jobs created)
Year 1	0	15	15
Year 2	0	30	30
Year 2	0	30	30
Year 4	0	30	30
Year 5	0	30	30
Total	0	30	30
Total number of crediting years	5		

Annual average over the crediting period	0	27	27
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SDG 13 – Climate Action

Climate change mitigation. Reduction in GHG emissions compared to the baseline scenario.

	Baseline estimate (t CO ₂ e)	Project estimate (t CO ₂ e)	Net benefit (t CO ₂ e)
PER SOLAR LIGHT	0.092	0	0.092

Overall Project	Baseline estimate (t CO ₂ e)	Project estimate (t CO ₂ e)	Net benefit (t CO ₂ e)
Year 1	9,198	0	9,198
Year 2	32,193	0	32,193
Year 2	59,971	0	59,971
Year 4	58,131	0	58,131
Year 5	53,532	0	53,532
Total	213,026	0	213,026
Total number of crediting years	5	5	5
Annual average over the crediting period	42,605	0	42,605

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 1.1 , SDG 3.9 and SDG 13.2

Data / Parameter	OF _y
Unit	Fraction
Description	The percentage of project lamps distributed to end users that are operating and in service
Source of data	Ex post monitoring household survey done 3 rd year.
Value(s) applied	100% years 1-3 and 80% years 4-7 (actual value to be based on household survey for years 4-7)
Measurement methods and procedures	For project lamps that will claim emission reductions for up to seven years, ex post monitoring surveys to determine percentage of project lamps distributed to end users that are operating and in service shall be conducted during the third year of the crediting period. While the percentage of project lamps that are operating and in service can be assumed to equal 100 per cent in year 1, 2, and 3, the result of ex post monitoring survey undertaken during the third year shall be used in years 4, 5, 6 and 7, as per paragraph 32. Only project lamps with a unique project marking (per paragraphs 19(f) or 38) can be counted as operating and in service. While project lamps replaced as part of a regular maintenance or warranty program can be counted as operating, project lamps cannot be replaced as part of the survey process and then counted as operating. The above ex post monitoring sampling surveys shall be conducted for each batch of project lamps. Alternatively, the result of a sampling survey of the first batch may be used as a proxy to subsequent batches (e.g. the operating rate in year 4 for the project lamps installed in year 1 could be used for the operating rate in year 5 for the project lamps installed in year 2).
Monitoring frequency	Survey conducted during the third year of the crediting period

QA/QC procedures	The survey results will be reviewed by the Project Management team to conduct QA/QC on the data entry, data processing and completeness and approve the survey data used for the SDG calculations.
Purpose of data	Used to determine the total number of operational Project lamps to calculate the Project's emission reductions and expenditure savings in years 4, 5, 6 and 7.
Additional comment	

SDG13

Data/parameter	GF _y
Unit	Fraction
Description	Grid factor in year y
Source of data	AMS-III.AR V.7.0
Value(s) applied	1.0
Measurement methods and procedures	Default value for GF _y is 1.0 as defined in AMS-III.AR V.7.0 under charging option 3a. The Project Solar Light is charged by a renewable energy system included as part of the project lamp (i.e. a photovoltaic system) . The default value will remain constant during the crediting period.
Monitoring Frequency	Annual
QA/QC procedures	-
Purpose of data	Supports calculation of Baseline Emissions per project lamp per year (tonnes CO ₂ e), BE _y
Additional comment	

SDG13

Data/parameter	DB _y
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Unit	Fraction
Description	Dynamic Baseline Factor (change in baseline fuel, fuel use rate, and/or utilization during crediting period in year y
Source of data	AMS-III.AR V.7.0
Value(s) applied	1.0
Measurement methods and procedures	Option 1: default of 1.0 in the absence of relevant information
Monitoring Frequency	Annual
QA/QC procedures	-
Purpose of data	Supports calculation of Baseline Emissions per project lamp per year (tonnes CO ₂ e), Bey
Additional comment	

SGD 1.1

Data / Parameter	$Ex_{h,p,\$}$
Unit	USD
Description	Annual expenditure on lighting per household in the project scenario
Source of data	Ex post monitoring household survey
Value(s) applied	99.18 (actual value to be based on household SDG 1 survey)
Measurement methods and procedures	To be based on answers from surveyed households and recorded in a Project Household Survey form provided by NSL. The value used will be the average value for this parameter from all the surveyed households. The data captured will include the average weekly expenditure in candles, kerosene and batteries for lighting per household, and converted into an annual value (multiplied by 52 weeks per year). The Local Data Management Partner will send an electronic copy of all

	surveys to the Project Developer for tabulation in project SDG calculation spreadsheet.
Monitoring frequency	Once during the crediting period. Value fixed afterwards for subsequent years
QA/QC procedures	The survey results will be reviewed by the Project Management team to conduct QA/QC on the data entry, data processing and completeness and approve the survey data used for the SDG calculations.
Purpose of data	To calculate Ex_{sl} (the annual financial savings per household due to a reduction in expenditures for lighting)
Additional comment	

SDG 1.1.1/ SDG 3.9 / SDG 7.1.1 / SDG 13.2

Data / Parameter	N					
Unit	Units					
Description	total number of project solar lights distributed during the crediting period					
Source of data	Project Database					
Value(s) applied	Number of solar lights distributed	Year 1	Year 2	Year 3	Year 4	Year 5
	Solar lights distributed	100,000	250,000	302,000	0	0
	Cumulative lights distributed	100,000	350,000	652,000	652,000	652,000
Measurement methods and procedures	Measurements to be based on NSL records kept following distribution and sales transactions of project solar lights. The data will include unique solar light serial number, user name, contact details and date of purchase/deployment will be recorded at point of sale using the Project app and then transmitted to Project Database digitally.					
Monitoring frequency	Quarterly and aggregated on a yearly basis					

QA/QC procedures	The Local Project Representatives and sales agents will be trained in the recording and use of the data required for the project at the point of sale, repair or warranty of the Project lamps. The Project Management team will check on a regular basis the digital data stored in the Project Database and electronic records will be kept for the crediting period + 2 years.
Purpose of data	Used for calculating baseline emissions for SDG 13 and access to clean lighting per SDG7. Used for calculation of sample survey for SDG 1 and SDG 3.
Additional comment	

SDG 4.4

Data / Parameter	Training
Unit	Number
Description	Training sessions for employees per year
Source of data	Employment records
Value(s) applied	1
Measurement methods and procedures	Recording the number of training sessions for employees (male / female) in the VPA for distribution, after sales, monitoring & management
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	SDG 4 contribution
Additional comment	-

SDG 3.9

Data / Parameter	Health and safety
Unit	% HH experiencing fewer respiratory diseases after adopting solar lighting
Description	Number of households visited medical facilities/dispensary for treatment of respiratory issues

	etc. such as cough, shortness in breath, pneumonia and other respiratory issues
Source of data	Ex post monitoring household survey
Value(s) applied	80%
Measurement methods and procedures	To be based on answers from surveyed sampled households and recorded in a SDG1/ SDG3 survey provided by NSL. Estimation based on survey conducted among a sample of beneficiaries. The data captured will include the end-users perceptions of air quality in their household, as well as their reported incidences of a reduction in common symptoms of poor air quality such as itchy eyes. The Local Data Management Partner will send an electronic copy of all surveys to the Project Developer for tabulation in project SDG calculation spreadsheet.
Monitoring frequency	
QA/QC procedures	Once during the crediting period. Value fixed afterwards for subsequent years
Purpose of data	The survey results will be reviewed by the Project Management team to conduct QA/QC on the data entry, data processing and completeness and approve the survey data used for the SDG calculations
Additional comment	SDG 3 contribution

SDG 8.5

Data / Parameter	Employment
Unit	Number
Description	Employment and income generation
Source of data	Employment records
Value(s) applied	30
Measurement methods and procedures	Recording the number of employees (male / female) in the VPA for distribution, after sales, monitoring & management

Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	SDG 8 contribution
Additional comment	-

Safeguarding Principle 9.4 Release of pollutants

Data / Parameter	E-Waste
Unit	Project Lamps
Description	Number of units/parts sent for safe disposal or recycling.
Source of data	Project database/Receipts from E-waste organisations
Value(s) applied	1
Measurement methods and procedures	Recording the number units sent for safe e-waste disposal and recycling.
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	E-Waste and recycling monitoring
Additional comment	-

B.7.2. Sampling plan

The Project will conduct two sets of sample surveys, each one for the following objective:

- (1) To determine the expenditures in lighting by households during the project scenario (Data Unit: $Ex_{y,h,p}$). To determine reported incidences of eye infections and respiratory diseases as part of the project scenario. This is referred later in this section as the "SDG1/SDG3 survey".
- (2) To determine the percentage of lamps (Data Unit: Fraction) that are still operational during the 3rd year of the crediting period with a 90/10 confidence/precision (as per methodology AMS-III.AR Version 7.0). This is referred later in this section as the "Operational Lamps Survey".

Each one of the two sample survey plans is described below.

SDG 1/ SDG 3 Survey

Field Measurement Objectives and Data to be collected: A survey will be carried out using a closed questionnaire. The questionnaire will be carried out during the first two years of the crediting period following the deployment of the applicable solar lights. The results of this survey will allow NSL to calculate the USD value for average household savings and the calculated % of household visited that report experiencing fewer respiratory diseases after adopting solar lighting. The following questionnaire is an example of some of the questions to be included in the SDG 1/ SDG 3 Survey:

Household Information	• User name • Contact details
Information on Project Solar Light	• Serial number • Date of solar light purchase/deployment • Local Distribution Partner involved in sale
Questions regarding expenditures for lighting	• What does the household use for lighting? • What is the primary method of lighting? • How much does the household spends on lighting (candles, kerosene, batteries) on average per week in Dollars (USD)?
Questions regarding air quality	• Do they feel the household air quality has improved since purchasing a solar light? • Are they reporting less incidences of itchy eyes since reducing/stopping their use of fossil fuel forms of lighting? • Are they reporting less incidences of respiratory disease since reducing/stopping their use of fossil fuel forms of lighting?
Situational check	• Is the solar light still in use? • How many solar lights they own?

Target Population:

The Target Population will be households in Zimbabwe that have purchased project solar lights deployed under this VPA.

Operational Lamps Survey

Field Measurement Objectives and Data to be collected: A survey will be carried out using a closed questionnaire. The questionnaire will be carried out during the 3rd crediting year following the deployment of the applicable solar lights. In addition to the questionnaire, the Local Data Management Partner will also physically check

whether the household is in possession of the lamps and whether the lamps work. The following questionnaire will be used for the survey:

Household Information	• Username • Contact details
Information on Project Solar Light	• Serial number • Date of solar light purchase/deployment • Local Distribution Partner involved in sale
Physical check of solar light	• Is the household in possession of 1 or more solar lights? – picture required. • Does the solar light deployed serial number match the Project Database? – picture required. • Does the solar light work? - picture required.
Situational check	• Is the solar light still in use?

Target Population:

The Target Population will be households in Zimbabwe that have purchased project solar lights deployed under this Project and are in year 3 of use following purchase/deployment.

Sampling Process

The sampling process is defined by a sampling frame and a sampling plan.

- The sampling frame applies to both the SDG1/SDG3 Survey and the Operational Lamp Survey.
- The sampling plans for SDG1/SDG3 Survey and the Operational Lamp Survey are not the same, as described in the sections below.

The Sampling Frame:

Will consist of the sales records that are stored in the Project Database with information on the households that have purchased or received solar lights. The sampling will be based on the serial number of the Project Lamps.

The Sampling Plan:

The sampling will be made as follows:

The Operational Lamp Survey needs to estimate OF_y , the percentage of lamps (Data Unit: Fraction) that are still operational during the 3rd year of the crediting period.

The SDG1/SDG3 Survey needs to estimate $Ex_{h,p,\$}$, the annual expenditure on lighting per household in the project scenario (Data Unit: Zimbabwe Dollar ZWL)

In accordance with paragraph 37 of the methodology, the following survey principles shall be followed for activities related to determining number of project lamps in service and operating under the project:

- The sampling size is determined by minimum 90 per cent confidence interval and the 10 per cent maximum error margin; the size of the sample shall be no less than 100;
- Sampling must be statistically robust and relevant, i.e. the survey has a random distribution and is representative of the target population (size, location);
- The method to select respondents for interviews is random;
- The survey is conducted by site visits;
- Only persons over age 12 are interviewed.

Individual participants will be randomly selected from the project database. Sample sizes will be in line with the Gold Standard and CDM methodology requirements. The random sample group is reselected for every monitoring period to ensure the selection remains random.

The sample size will be determined as per the CDM guidelines and the minimum total sample size will be 100, as per required by the methodology AMS-III.A.R. version 7.0.

The surveys will be conducted so as to ensure that they are within the end date of the respective monitoring periods for each VPA.

B.7.3. Other elements of monitoring plan

- Operational and management structure

The CME has overall responsibility for the implementation, control and review of the monitoring plan. All monitored data will be submitted to the CME for review.

The VPA Implementer has responsibility for the actual implementation of the monitoring plan at VPA level and will employ and train staff to undertake monitoring activities accordingly.

- Provisions for data archiving

All monitoring data will be collected by the project team and submitted to the CME for QA/QC. Data will be archived by the CME electronically and securely on its own server for future auditing purposes.

- Responsibilities and arrangements for data collection

The CME will provide the format and means for the collection of monitoring data – this may be electronically collected or via a paper-based system, or a combination of both. The project team will collect monitoring data in the first instance by appointing by trained field-based monitoring teams, which may also be a 3rd party. All data collected in the field will be retained securely for future audit purposes.

Once collected data will be collated and screened for QA/QC purposes by the CME for further QA/QC.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project.

The start date of the project is 01/03/2022 which is the date when the first Project units were distributed as part of this VPA.

C.1.2. Expected operational lifetime of project.

15 years

C.2. Crediting period of project

C.2.1. Start date of crediting period.

01/03/2023

C.2.2. Total length of crediting period

5 years twice renewable

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored.

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3. Community Health, Safety and Working Conditions The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	The Project Developer will implement the solar light deployment in the country based on advice from the Government and a COVID-19 safe protocol to ensure schools and other distribution centres are re-opening. The situation will be monitored to avoid the project becoming a vector for transmission.
Principle 9.4 Release of pollutants	The Project Lamps have a useful life of 7 years. In the event of a solar light failing the light is designed on a plug & play architecture that enables easy, field replacement of failed component and recovery of failed component for recycling by the sales agent network. The components or solar lights at the end of life will be collected and disposed or recycled safely according to the Host Country's regulations and available facilities. At end of life, the project intends to sell a replacement solar light and recovery the end of life product for recycling via the sales agent network and adhering to local country regulations on recycling and disposal of waste.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project aims to generate livelihood benefits equally for both men and women, which is in line with the Gold Standard foundational gender sensitive requirement of 'do no harm' as stated point 4.2 of the Gold Standard Gender Policy. In compliance with this principle, safeguards and the stakeholder consultations will be gender-sensitive as shown in the stakeholder consultation plan.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The revised National Gender Policy (NGP) was unveiled on July 6, 2017 with a call for gender justice, equality, integration, inclusiveness, and shared responsibility for sustainable development in Zimbabwe. Developed through a consultative process and coordinated by the Ministry of Women's Affairs, Gender and Community Development with support from UNDP and UN Women, the NGP provides a broad framework to guide and coordinate all efforts for addressing gender inequality and discrimination in Zimbabwe. This NGP's vision is translated into 10 priority thematic areas namely: Gender, Constitutional and Legal Rights; Gender and Economic Empowerment; Gender, Politics and Decision Making; Gender and Health; Gender, Education and Training; Gender Based Violence; Gender, Environment and Climate Change; Gender, Media and Information Communication Technology (ICTS); Gender and Disability; and Gender, Culture and Region²¹. A core purpose of this project is to support the deployment of solar lights equally across genders through schools to facilitate increased levels of study and grade attainment by pupils that is currently constrained by the significant use of fossil fuel-based sources of lighting. This complements the policy of gender, education, and training by addressing their indicator of retention rate of girls in secondary schools by increasing the ability of girls to

²¹

<https://www.zw.undp.org/content/zimbabwe/en/home/presscenter/articles/2017/07/06/milestones-as-new-national-gender-policy-is-launched0.html>

be able to study at night, by providing them with high quality, renewable lighting.

Additionally, the projects will provide economic opportunities for women, such as roles in product promotion, which will complement the governments priority area of gender and economic empowerment by increasing the proportion of the population of women participating in the economy, which sat at only 60% in 2016 as recorded by the FAO²². This project will therefore contribute towards an increase the balance in income distribution between men and women, of which both are measurement indicators set out in the policy.

This project is also aligned with Zimbabwe's broader gender strategies by incorporating women in its decision-making processes. The Zimbabwe Gender Commission has outlined dialogue and decision-making power as two of its main strategies to achieve gender equality²³. The commission promotes the inclusion of women in decision-making settings, as well as active dialogue between genders. Moreover, the Regional Strategy and Framework of Action for Addressing Gender Based Violence²⁴, published by the Southern African Development Community – which Zimbabwe is a part of – states that member countries should address the underlying causes for gender-based violence. These include the lack of female voices in decision-making, unyielding gender stereotypes and barriers to female growth. Namene's project is in line with these guidelines. While the lights facilitate the education of girls and young women, the stakeholder consultation process will hear the voices of the community. This part of the project aims to achieve as much of an even split as culturally possible between female and male stakeholders.

²² "National gender profile of agriculture and rural livelihoods: Zimbabwe | [FAO](https://www.fao.org/)". www.fao.org. Retrieved 25 October 2018.

²³ <https://zgc.co.zw/download/1342/>

²⁴ https://www.sadc.int/files/8815/6397/5761/SADC_Regional_Strategy_and_Framework_for_Action_on_GBV_ENGLISH.pdf

	Consultants will seek out and listen to the female voices in visited communities. These voices will be translated into valuable information, which will inform future decisions made by Namene regarding those communities. The nature of the project is highly distributed and access to households is facilitated via schools. Distribution of the solar lights is dependent on demographic mix of male and females as the aim of the project is to distribute solar lights to all rural households and thereby school children by making the solar lights highly affordable. The usage of a solar light does not create any discriminatory risk. The distribution of solar lights may potentially reduce workload for females associated with the avoided travel to purchase kerosene and/or paraffin candles and alleviate the emotional burden regarding fire risk in the household. No risks or hazards are introduced through the usage of solar lights. Reduces risks of fires and supports female safety in the dark evenings if walking around at night.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No – The preliminary assessment of the project shows no requirement for an expert. The project intends to distribute at schools and/or local health centres, where gender is not a barrier to access.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No – The stakeholder consultation plan shows that the consultations are available to all people that wish to participate.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

As part of the stakeholder consultation, three consultations were held in the three provinces targeted in Zimbabwe within the project boundary. These consultations were held in the districts of Chinhoyi (30/08/2021), Marondera (01/09/2021) and Masvingo (03/09/2021). Following the format provided by “stakeholder consultation and engagement requirements version 1.2” and “stakeholder consultation report template guide v1.1.” Following the physical consultations, a feedback round took place between 21/11/2021 and 20/01/2022.

Stakeholders were invited by varying invitation methods that were tailored to different stakeholders. Namene’s local team and partners advised on the language and tone of invitations based on their cultural knowledge. The on-site teams also advised on the use of physical letters to stakeholder categories A and B, and the use of emails for categories C and D. These are justified by the stakeholders’ communication needs. Categories A and B are mostly based in rural off-grid regions where email is rarely used. Therefore, the primary means of communication to co-ordinate and invite these stakeholders to meetings was a physical letter or verbal explanations. Conversely, the categories C and D have more access to the internet, which favoured the use of instant communication.

Details about invitation and attendance per gender can be found below:

Invitation	Men	Women
Chinhoyi	19	11
Marondera	19	12
Masvingo	21	8

TOTAL	59	31
Percentage	65	35
Attendance	Men	Women
Chinhoyi	10	7
Marondera	17	4
Masvingo	19	10
TOTAL	46	21
Percentage	68	32

The comments received were mostly positive regarding the project aim and potential benefits for the distribution of solar lights, leading to savings in households, reduced indoor air pollution, increased hours of study and reading, as well as potential opportunities for local employment.

The stakeholders voiced concerns on issues that have been considered in our safeguarding principles, such as E-waste measures. These points will be emphasized and explained in further details for clarity to end users by agents as the project moves forward into awareness campaigns and distribution.

Other comments were taking for consideration, such as potentially monitoring the improvement in grades of school children and other economic benefits. These will depend on the feasibility of accessing this data and availability of data by the corresponding authorities and assess whether there is a direct correlation that can be attributed to the project activity. No further changes are expected to be implemented based on the stakeholder consultation process.

E.2. Final continuous input / grievance mechanism

	Method Chosen (include all known details e.g. location of book, phone, number,	Justification of Choice (best practice)
--	---	--

	identity of mediator)	
Continuous Input / Grievance Expression Process Book (mandatory)	The grievance books will be accessed through village schools, headmen and churches. Exact locations will be confirmed by Namene on a local basis, based off the needs of the community	Based of community feedback in the meetings, these were the locations suggested. They are appropriate as they are publicly available and can be placed under the care of a responsible member of the local community, such as a headteacher.
GS Contact (mandatory)	help@goldstandard.org	-
Telephone access (optional)	A hotline will be provided by a local partner during distribution	A local number will allow end users to contact to express grievances with more ease
Internet/email access (optional)	Climate@namenesolar.com	This email reaches the manager of this project as well as the larger climate department at Namene Solar.
Nominated Independent Mediator (optional)	-	-
Other	-	-

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No	The Project complies with internationally proclaimed human rights and shall not be complicit in violence or human rights abuses.	N/A
2. The Project shall not discriminate with regards to participation and inclusion		The Project does not discriminate with regards to participation and inclusion given the broad reach of the project across all stakeholders.	N/A

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		The Local Distribution Partner will be required to comply with Namene Solar Ltd's policies on Human Rights.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks	No	The revised National Gender Policy (NGP) was unveiled on July 6, 2017 with a call for gender justice, equality, integration, inclusiveness, and shared responsibility for sustainable development in Zimbabwe. Developed through a consultative process and coordinated by the Ministry of Women's Affairs, Gender and Community Development with support from UNDP and UN Women, the NGP provides a broad framework to guide and coordinate all efforts for addressing gender inequality	N/A

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		and discrimination in Zimbabwe. This NGP's vision is translated into 10 priority thematic areas namely: Gender, Constitutional and Legal Rights; Gender and Economic Empowerment; Gender, Politics and Decision Making; Gender and Health; Gender, Education and Training; Gender Based Violence; Gender, Environment and Climate Change; Gender, Media and Information Communication Technology (ICTS); Gender and Disability; and Gender, Culture and Region ²⁵ .	

²⁵ <https://www.zw.undp.org/content/zimbabwe/en/home/presscenter/articles/2017/07/06/milestone-as-new-national-gender-policy-is-launched0.html>

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		A core purpose of this project is to support the deployment of solar lights equally across genders through schools to facilitate increased levels of study and grade attainment by pupils that is currently constrained by the significant use of fossil fuel-based sources of lighting. This complements the policy of gender, education, and training by addressing their indicator of retention rate of girls in secondary schools by increasing the ability of girls to be able to study at night, by providing them with high quality, renewable lighting. Additionally, the projects will provide economic opportunities for women, such as roles in product promotion,	

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		which will complement the governments priority area of gender and economic empowerment by increasing the proportion of the population of women participating in the economy, which sat at only 60% in 2016 as recorded by the FAO²⁶. This project will therefore contribute towards an increase the balance in income distribution between men and women, of which both are measurement indicators set out in the policy. This project is also aligned with Zimbabwe's broader gender strategies by	

²⁶ "National gender profile of agriculture and rural livelihoods: Zimbabwe | [FAO](http://www.fao.org)". www.fao.org. Retrieved 25 October 2018.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		incorporating women in its decision-making processes. The Zimbabwe Gender Commission has outlined dialogue and decision-making power as two of its main strategies to achieve gender equality ²⁷ . The commission promotes the inclusion of women in decision-making settings, as well as active dialogue between genders. Moreover, the Regional Strategy and Framework of Action for Addressing Gender Based Violence ²⁸ , published by the Southern African Development Community – which Zimbabwe is a part of – states that member countries	

²⁷ <https://zgc.co.zw/download/1342/>

²⁸ https://www.sadc.int/files/8815/6397/5761/SADC_Regional_Strategy_and_Framework_for_Action_on_GBV_ENGLISH.pdf

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		should address the underlying causes for gender-based violence. These include the lack of female voices in decision-making, unyielding gender stereotypes and barriers to female growth. Namene's project is in line with these guidelines. While the lights facilitate the education of girls and young women, the stakeholder consultation process will hear the voices of the community. This part of the project aims to achieve as much of an even split as culturally possible between female and male stakeholders. Consultants will seek out and listen to the female voices in visited communities. These voices will be translated into valuable information, which	

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		will inform future decisions made by Namene regarding those communities. The nature of the project is highly distributed and access to households is facilitated via schools. Distribution of the solar lights is dependent on demographic mix of male and females as the aim of the project is to distribute solar lights to all rural households and thereby school children by making the solar lights highly affordable. The usage of a solar light does not create any discriminatory risk. The distribution of solar lights may potentially reduce workload for females associated with the avoided travel to purchase kerosene	

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		and/or paraffin candles and alleviate the emotional burden regarding fire risk in the household. No risks or hazards are introduced through the usage of solar lights. Reduces risks of fires and supports female safety in the dark evenings if walking around at night.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Potentially	The Project will improve Stakeholders health by removing exposure to dangerous and polluting kerosene lamps, candles and dry cell batteries. The Project solar lights could also improve safety by reducing the risks of fires in households and	The Project Developer will implement the solar light deployment in the country based on advice from the Government and a COVID-19 safe protocol to ensure schools and other distribution centres are re-opening. The situation will be monitored to

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		improve road safety when using the light at night while cycling or walking on the streets. Covid-19 transmission as a result of activities to deploy solar lights.	avoid the Project becoming a vector for transmission.
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	Not relevant to the project or Context	N/A
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples	No	Not relevant to the project or Context	N/A

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
(temporary or permanent, full or partial)?			
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	Not relevant to the project or Context	N/A
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	Not relevant to the project or Context. The Project does not involve any land tenure or other land rights.	N/A
>>			
Principle 4.4 - Indigenous People			

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	Not relevant to the project or Context. The government of Zimbabwe does not recognise any specific group as indigenous to the country ²⁹ . Despite this, NSLC understands the claims of some groups and will be sensitive of the issue.	N/A
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	All employees of NSLC, contracts that NSLC enters into as a counter-party and day-to-day activities follow strict rules on anti-corruption. A copy of these rules is available on request. Local Distribution Partner will be required to implement	N/A

²⁹ [Zimbabwe - IWGIA - International Work Group for Indigenous Affairs](#)

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		company policies on anti-corruption.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include:	No	The Project will adhere to labour laws and requirements. The Project Developer has internal policies in place to adhere to these principles and will require to the Local Distribution Partner to follow these principles as well. The Project will not involve any child labour. The Project Developer will enter into agreements with the Local Distribution Partners that include child labour policies.	N/A

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for		The Project Developer and its Local Distribution Partner will ensure that workers will have the adequate training and they report accidents or incidents.	

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation? >>	No	The Project does not cause negative economic consequences during and after the project as a key objective is to decrease poverty by avoiding the cost of kerosene and paraffin candles. In addition, the solar	N/A

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		lights will be sold at an affordable price to users (determined during Stakeholder consultations) to economically support a local sales and after-sales support channel. Negative economic consequences were not raised as a risk during Stakeholder Consultation rounds	
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	Project adopts Methodology AMS-III.AR Version 7.0 that states: The Project emissions from the distribution of solar lights, P _{Ey} , are zero as the charging option defined in paragraph 3(a) is applicable to the Project Lamps.	N/A
>>			
Principle 7.2 Energy Supply			

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The Project will use rechargeable solar lights that are autonomous in operation.	N/A
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	Not relevant to the project or context. The project will not have any impact or interaction with water or water bodies	N/A
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional	No	Not relevant to the project or context. The project will not	N/A

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
erosion and/or water body instability or disrupt the natural pattern of erosion?		have any impact or interaction with water or water bodies	
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	Not Relevant to project or context	N/A
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	NO	Not relevant to project or context	N/A
>>			
Principle 9.3 Genetic Resources			

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	Not relevant to project or context	N/A
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	Potentially	The Project Lamps has a bill-of-materials that includes UV resistant durable plastic casing, metal stand, printed circuit board (PCB), solar PV panel and a rechargeable LiFePO4 battery. All internal components (PV panel, PCB and battery are configured as plug&play to enable field repair and easy	The Project Lamps have a useful life of 7 years. In the event of a solar light failing the light is designed on a plug & play architecture that enables easy, field replacement of failed component and recovery of failed component for recycling by the sales agent network. The components or solar lights at the end of life will be
>>			

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
		separation for end-of-life recycling and disposal. Exposure to risk of pollutant release is at end-of-life of product or from defect parts that are replaced for field repairs	collected and disposed or recycled safely according to the Host Country’s regulations and available facilities. At end of life, the project intends to sell a replacement solar light and recovery the end of life product for recycling via the sales agent network and adhering to local country regulations on recycling and disposal of waste.
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project Lamps bill-of-materials does not make use of materials or chemicals that are subject to international bans or phase-outs.	Same as per the comment above
>>			
Principle 9.6 Pesticides & Fertilisers			

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Will the Project involve the application of pesticides and/or fertilisers?	No	Not relevant to the project or context	N/A
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	Not relevant to the project or context	N/A
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	Not relevant to the project or context	N/A
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	Not relevant to the project or context	N/A
>>			

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	Not relevant to the project or context	N/A
>>			
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be	No	Not relevant to the project or context	N/A

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
present through transboundary affects?			
> >			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	Namene Solar Lights Limited
Registration number with relevant authority	11318851 -
Street/P.O. Box	Holland Street 51
Building	
City	London
State/Region	N/A
Postcode	W8 7JB
Country	England
Telephone	+49 162 138 2414
E-mail	climate@namenensolar.com
Website	www.namenensolar.com
Contact person	Bernardo Lazo
Title	Managing Director Namene Solar Lights
Salutation	Mr
Last name	Lazo
Middle name	
First name	Bernardo
Department	
Mobile	
Direct tel.	
Personal e-mail	Bernardo.Lazo@namenensolar.com

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Annex A of [Principles and Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.1	7 October 2020	- Hyperlinked section summary to enable quick access to key sections - Improved clarity on Key Project Information - Inclusion criteria table added - Gender sensitive requirements added - Prior consideration (1 yr rule) and Ongoing Financial Need added - Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity - Improved Clarity on SDG contribution/SDG Impact term used throughout - Clarity on Stakeholder Consultation information required - Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption